

Macroeconomic Surprises, DTW Price Patterns and the Intraday Response of E-mini S&P 500 (ES), E-mini Nasdaq-100 (NQ) and 10-Year Treasury Note (ZN) Futures

A walk-forward study of the macro surprise, the pre-event market state, and Dynamic Time Warping (DTW) price patterns

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Project: BlackRock industry project

September 2026

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Abstract

Scheduled US macroeconomic releases (CPI, payrolls, PCE, retail sales, ISM, and others) are the most predictable sources of intraday volatility in futures markets: the release time is known in advance, and the market's expectation is published as a consensus forecast. This study asks whether a trader who reacts **after** a release, and only with information available at that moment, can profit from the market's response net of realistic costs, in three futures markets: the **E-mini S&P 500 (ES)**, the **E-mini Nasdaq-100 (NQ)** and the **10-year Treasury note (ZN)**.

Two families of signals are studied: the **macro surprise** (actual minus consensus, standardised causally) and the **Dynamic Time Warping (DTW) price pattern**, which matches the shape of the market's price path around a release to similar historical releases. A third ingredient, the **pre-event market state** (volatility, momentum, drawdown, overnight move), is tested as a conditioning variable.

The main findings are:

1. **CPI is the release that matters.** A one-standard-deviation hotter headline CPI print lowers ES by about **18-24 bps** and NQ by about **28 bps**, with R^2 of 0.27-0.30 at five minutes. The

same print lowers ZN by about **10 bps**. Other releases move markets but carry much less directional information.

2. **Most of the reaction happens in the first minute**, which no strategy can capture. In equity-index futures a **tradeable drift** remains for about **30 minutes** (about 8–9 bps per 1-SD CPI surprise); in ZN the drift is essentially zero (0.8 bps per 1-SD).
3. **A simple surprise-sign strategy is profitable after costs in equity futures**. Entering one minute after the release and holding for 30 minutes, the CPI strategy earns a walk-forward net Sharpe of about **1.0 on ES** and **0.8–1.0 on NQ** (2019–2026). Extending it to about ten screened releases raises the trade count from ~10 to **~100 trades a year per market** at a net Sharpe of **0.65 (ES)** and **0.71 (NQ)**; the ES+NQ book reaches **~195 trades a year at Sharpe 0.72**.
4. **The edge is regime-dependent**. About 75–80% of CPI profits come from the 2022 inflation shock; the multi-release book depends less on 2022 (Sharpe ex-2022 0.28 on ES, 0.44 on NQ).
5. **Neither the pre-event state nor DTW improves the direction out of sample**. The state strongly predicts the *size* of the move, but not its direction. DTW has no directional information at an executable entry in any market or any of five post-release windows.
6. **ZN is not tradeable with this approach**. The bond market prices surprises almost entirely in the first minute; every ZN strategy loses money at a cost of about 3 bps per round trip.
7. **At equal costs, the surprise strategy is the most robust approach in the team**. Repricing all four workstreams under the same costs and window shows that high-turnover DTW strategies are positive only at very low costs, and that part of a teammate’s reported performance came from an entry at the pre-release price.

1. Introduction and research questions

1.1 Motivation

Macroeconomic releases are the scheduled “information shocks” of financial markets. At 08:30 ET on a CPI day, the market learns in one second whether inflation was higher or lower than expected, and futures prices jump. The size and direction of that jump depend on the **surprise**, the difference between the published number and what the market expected (the consensus of economists’ forecasts surveyed by Bloomberg).

For a trader, two facts make this interesting and difficult at the same time:

- **The direction is predictable in principle**. Economic theory says what a surprise should do: a hot inflation print raises expected policy rates, which lowers equity valuations and bond prices.
- **The first move is not tradeable**. High-frequency participants reprice within milliseconds. A strategy that decides after seeing the number, with realistic latency, can only trade what happens **after** the first minute.

The project brief asked how the S&P 500 futures respond to macro surprises intraday, and whether that response depends on the market’s state before the release. During the project the question was widened to three markets (ES, NQ, ZN), to many releases rather than CPI alone, and to a second signal family (DTW price patterns) used by the rest of the team.

1.2 Research questions

#	Question	Where answered
RQ1	Which macro releases move ES, NQ and ZN, and by how much per unit of surprise?	Notebooks 02, 02b
RQ2	How much of the reaction happens in the first minute, and how much remains as a tradeable drift?	02b, 03, 03b

#	Question	Where answered
RQ3	Does the market's state before the release change the reaction to the same surprise?	03, 03b
RQ4	Can a trader profit from the surprise in real time, after costs, with only past information? Does conditioning on the state help?	04, 04b
RQ5	How many trades a year can a surprise strategy make across many releases, and does the edge survive?	05, 05b
RQ6	Does the DTW price pattern add information to the surprise? Can more post-release windows raise the trade count profitably?	06, 06b
RQ7	Does combining markets help, and how do the four team workstreams compare at equal costs?	07, 08

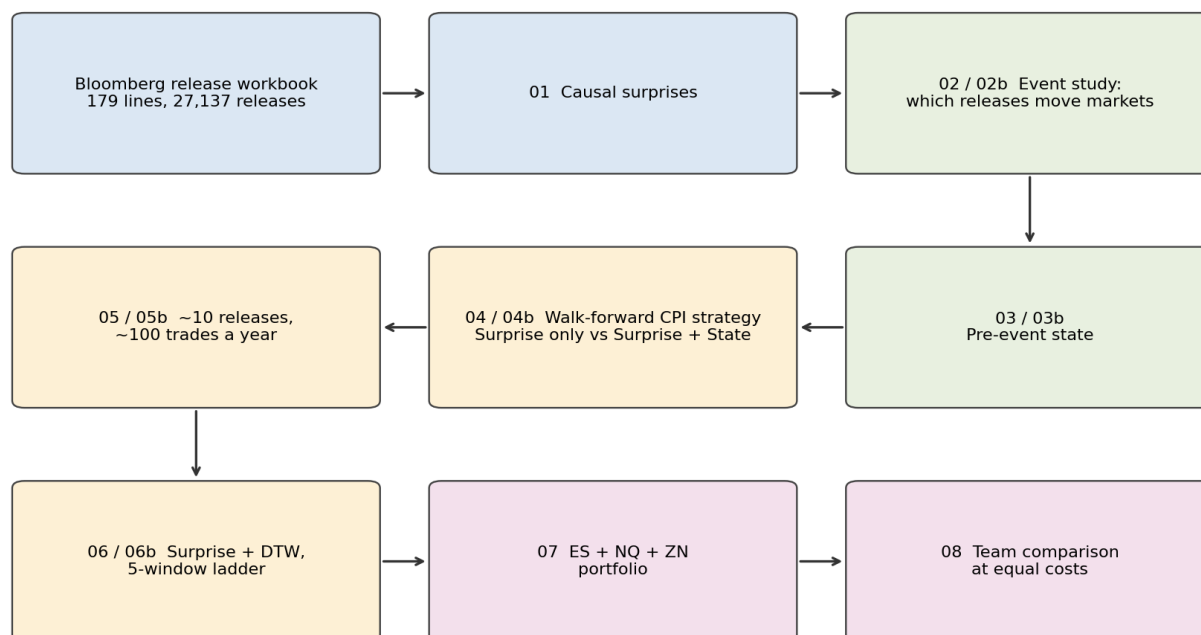
1.3 The project's three-way comparison

The original plan ends with a comparison of three model families, which this document follows throughout:

Model family	Idea	Notebooks
Surprise only	Trade in the direction implied by the surprise	04, 05
Surprise + State	Let the pre-event state change the size or sign of the trade	03, 04
Surprise + Pattern	Add the DTW price-pattern signal to the surprise	06

1.4 Pipeline overview

Market data: ES / NQ / ZN 1-minute bars → roll-adjusted log prices (used from Notebook 02 onwards)



Pipeline of the study.

2. The three markets

2.1 Why ES, NQ and ZN

	ES	NQ	ZN
Underlying	S&P 500 index	Nasdaq-100 index	10-year US Treasury note
Exchange	CME	CME	CBOT
Tick size	0.25 index points	0.25 index points	1/64 of a point
Value per point	\$50	\$20	\$1,000
Value of one tick	\$12.50	\$5.00	\$15.625
Typical price (2019–2026)	2,500–6,500	7,000–30,000	105–135
What moves it on a macro release	Growth, inflation, discount rates	Same as ES, more rate-sensitive (long-duration tech)	Expected policy rates and inflation
Round-trip cost used (2 ticks + \$4.50)	≈ 1.7 bps	≈ 0.7–0.8 bps	≈ 3.0 bps

ES is the most liquid equity-index future in the world and the project's main market. NQ is a second equity index with a higher share of long-duration technology stocks, so it should react **more** to inflation and rate news. ZN is a bond future: its price moves **opposite** to yields, so a hot inflation print or a strong jobs report should push ZN **down**. Together the three markets test whether the macro-surprise effect is a property of one contract, of equity indices, or of rate-sensitive markets in general.

2.2 The markets over the sample

The three markets: roll-adjusted daily price, 2010–2026



Figure 1. Roll-adjusted daily prices of ES, NQ and ZN, 2010–2026, rebased to 100. Grey: the walk-forward evaluation period (2019–2026). Red: the 2022 inflation shock.

The evaluation period covers very different regimes: the low-inflation years 2019–2020, the COVID crash (March 2020), the 2022 inflation shock and the fastest Fed tightening in forty years, and the disinflation of 2023–2026. Over 2019–2026, buying and holding one contract returned:

Buy and hold, 2019–2026	ES	NQ	ZN
Total return	+218%	+348%	−10.5%
Daily Sharpe	0.90	0.99	−0.23
Maximum drawdown	−29.5%	−38.3%	−23.6%

The strategies in this study hold positions for 30 minutes after a release, about 0.1–2% of the time, so they are **market-neutral** by construction (beta to buy-and-hold between -0.004 and $+0.006$ in every notebook). Buy-and-hold is shown only as a reference point.

2.3 What a release does to each market

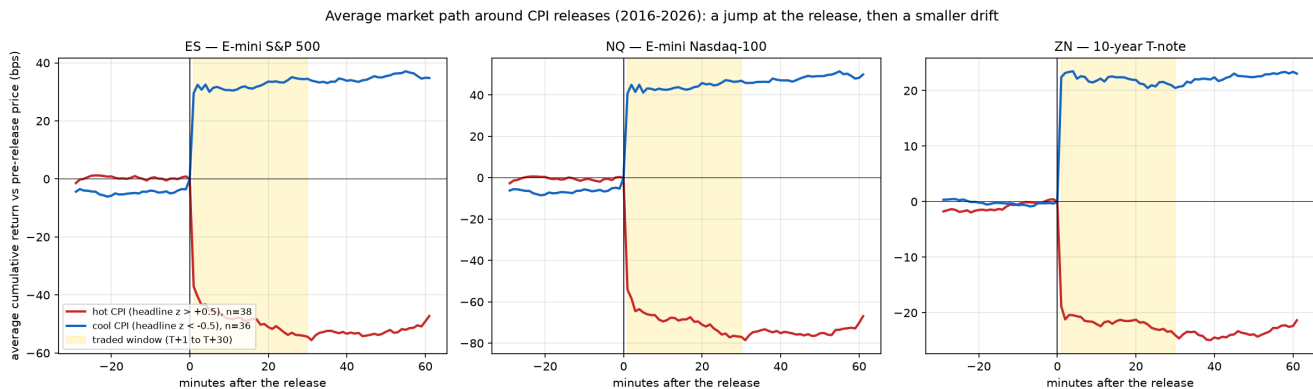


Figure 2. Average cumulative return around CPI releases, 2016–2026, relative to the last pre-release price, split into hot prints (headline surprise above $+0.5$ SD) and cool prints (below -0.5 SD). Yellow: the window a strategy can trade ($T+1$ to $T+30$ minutes).

Figure 2 summarises the whole project in one picture:

- **The jump.** At the release (minute 0) prices jump in the direction theory predicts. After a hot CPI print (38 releases with a headline surprise above $+0.5$ SD), ES falls about **37 bps**, NQ about **55 bps** and ZN about **19 bps** within the first minute. After a cool print (36 releases below -0.5 SD) they rise by about 30, 42 and 23 bps.
- **The drift.** In the equity indices the price keeps moving after the jump. After hot prints ES falls a further **~15–18 bps** (to about -55 bps at $T+30$) and NQ a further **~20 bps** (to about -77 bps). After cool prints the continuation is smaller (about $+4$ bps in both). This post-jump move is what a strategy that reacts one minute after the release can capture, and it is larger after bad news than after good news.
- **The difference between markets.** In ZN the continuation after the first minute is only about **4–5 bps** after hot prints and slightly **reverses** after cool prints. Relative to ZN's round-trip cost of about 3 bps, almost nothing is left to trade, which is why no ZN strategy survives costs (Sections 6.4–6.6).
- **Before the release** all three markets are flat on average: there is no systematic pre-positioning in the 30 minutes before CPI.

3. The two signal families: macro surprise and DTW

3.1 The macro surprise

For each release line i (for example “CPI MoM”) at release t , the raw surprise is

$$s_{i,t} = A_{i,t} - F_{i,t},$$

where A is the actual value and F the Bloomberg survey median. Raw surprises are not comparable across lines (payrolls are in thousands of jobs, CPI in tenths of a percent), so each is standardised **causally**, using only earlier releases of the same line:

$$z_{i,t} = \frac{s_{i,t}}{\text{sd}(s_{i,1}, \dots, s_{i,t-1})}, \quad \text{at least 24 previous releases,} \quad z \text{ capped at } \pm 5.$$

A line’s **sign** is estimated from history: a regression of the **instant jump** (the first-minute return) on z over earlier years tells whether a positive surprise has pushed that market up or down. Lines published at the same instant (for example the eight CPI lines) are grouped into a **release family**, and the family signal is

$$x_{f,t} = \text{mean}_{i \in f} (\text{sign}_i \times z_{i,t}),$$

so that $x > 0$ always means **good news for that market**. For ZN, “good news” means good for bond prices: a strong payrolls print is bad news for ZN but good news for ES.

Economic channel. A hot CPI print raises expected inflation and the expected path of the policy rate. Higher expected rates lower the present value of future earnings (bad for ES and especially NQ) and raise yields (bad for ZN). Strong payrolls raise expected growth (good for equities) but also expected rates (bad for bonds).

3.2 The DTW price pattern

Dynamic Time Warping measures the similarity between two time series that may be locally stretched or compressed in time. The team’s DTW strategy (Jack Duncan, Hoshea Zeng) uses it to answer: *“Have we seen a price reaction like this one before, and what happened next?”* The specification used here follows the peer strategy exactly:

Step	Specification
Query path	Log prices over the 31 minutes ending at the entry (T−30 to T+1), z-scored so that only the <i>shape</i> matters
Library	Every earlier release at the same clock time (for example 08:30), last 3 years , at most 600 most recent
Distance	DTW with a Sakoe-Chiba band of 5 minutes
Neighbours	The k = 15 nearest paths, weighted by $1/\text{distance} \times \text{recency}$ (2-year half-life)
Signal	Direction = weighted mean of the neighbours’ next-30-minute returns; dispersion = weighted standard deviation
Gate	Trade only when the direction is outside the middle 40% of its own previous-3-year range

A neighbour is used only if its own trade had already finished before the current release, so the signal is free of look-ahead. The code was validated on synthetic data in which the outcome depends on the path shape: it recovered the pattern with a rank IC of 0.52 and a 71% hit rate.

3.3 The pre-event state

Six **pre-registered** state variables describe the market just before the release, each with an economic hypothesis:

State	Definition (known before T)	Hypothesis
rv_20d	20-day realised volatility of daily closes	Higher uncertainty → larger reaction per unit of surprise
rv_pre_1d	Realised volatility of 5-minute returns over the last ~23 hours	Short-term stress amplifies the reaction
mom_20d	20-day log return	After strong rallies, bad news hits harder
dd_60d	Distance from the 60-day high	In drawdowns the market is more sensitive to inflation
overnight_ret	Previous 16:00 close → T-1	A large pre-release move means partial pre-positioning
prev_x	Previous surprise of the same release	Surprise streaks confirm a trend

Each state is converted to a percentile against a **trailing one-year** distribution (for intraday states, measured at the same clock time), so no full-sample information is used.

3.4 How the signals are combined into trades

Rule	Position
Naive sign	$\text{sign}(x)$, one contract, no estimation
Surprise model	$\hat{b}x$ with $\hat{b}$ re-estimated walk-forward; trade $\text{sign}(\hat{b}x)$ only if the absolute value of $\hat{b}x$ exceeds the round-trip cost
Surprise + State	$\hat{r} = a + bx + cs + d(x \times s)$, walk-forward
State filter	Surprise model, but skip the low-volatility tercile
Adaptive state	Each year, choose the state (or none) that worked best in the training data
Signal × volatility	Jack's sizing: position proportional to $x \times \text{recent volatility}$
Surprise + DTW agree	Trade the surprise only when DTW points the same way
Surprise + DTW blend	Average the two signs (opposite signs cancel)
DTW only	Sign of the DTW direction, after the gate

4. Data

4.1 Sources

Data	Source	Coverage	Used for
Macro releases	Bloomberg Economic Calendar workbook (Bloomberg Economic Releases.xlsx), 17 yearly sheets	2010-2026: 179 release lines, 27,137 release rows (release date and time, event name, survey median, actual, ticker)	Surprises, release calendar
ES 1-minute bars	Databento, continuous front-month ES.c.0 with instrument_id	July 2010 - June 2026	ES prices
NQ and ZN 1-minute bars	Jack Duncan's cleaned files fu-utures_1min_clean_v3_{NQ,ZN}.parquet	July 2010 - June 2026, 5.3-5.4 million bars each, 64 contract rolls	NQ and ZN prices
Team outputs (Notebook 08 only)	Hoshea Zeng's DTW factor file; Aaryen Mehta's walk-forward predictions (10, 30, 60 minutes)	2012-2026	Team comparison

The raw data are proprietary (Bloomberg, Databento) and are **not** included in the repository (.git-ignore excludes *.parquet and *.xlsx).

4.2 Samples

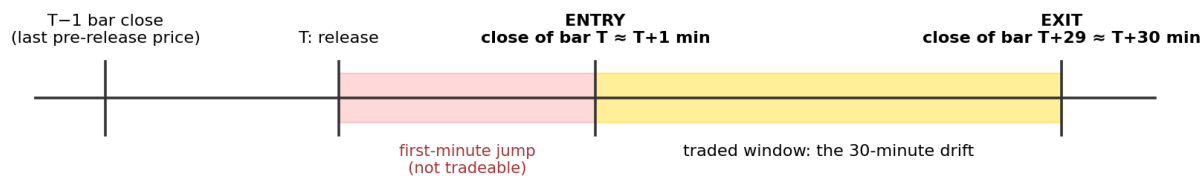
Sample	Size	Period
Core event study (CPI, NFP, PCE)	363 releases (122 CPI, 122 NFP, 119 PCE); 354 with NQ prices, 350 with ZN prices	2016-2026
Release families (lines grouped by timestamp)	143 lines with ≥ 30 releases $\rightarrow$ 75 families	2013-2026
Causal surprises	17,328 with a z-score (≥ 24 prior releases); 0.47% capped at $\pm 5\sigma$	2010-2026
Release timestamps with a price path	$\sim 6,100$ since 2016 ($\sim 5,850$ with a complete 30-minute trade return)	2016-2026
Walk-forward evaluation window	2019-01 to 2026-06 (7.5 years); 2016-2018 used only as history	

4.3 Data cleaning

- **Time zones.** Release times are Eastern Time and converted to UTC, with daylight-saving time handled explicitly; bars are in UTC.
- **Bar labels.** A 1-minute bar is labelled by its **start** minute: the bar labelled 08:30 covers 08:30:00-08:30:59 and closes at 08:31.
- **Contract rolls.** A continuous futures series jumps at each quarterly roll by the calendar spread (in NQ up to ± 700 bps). Returns across a change of contract are set to zero, which leaves returns within each contract exactly unchanged (checked: maximum difference 0).
- **Duplicates and missing values.** Releases are de-duplicated by (line, timestamp), keeping the last revision. A price is used only if a bar exists within 5 minutes of the required time.

5. Methodology

5.1 Timing of a trade



For a release at time T :

Quantity	Definition	Used in
Reaction over h minutes	$R_{T,h} = \ln P_{T+h-1} - \ln P_{T-1}$ (includes the jump)	Notebooks 02, 02b, 03
Jump	$\ln P_T - \ln P_{T-1}$ (close of the release bar vs the last pre-release close)	Line signs, impact screen
Drift / trade return	$D_{T,h} = \ln P_{T+h-1} - \ln P_T$ (entry about one minute after the release)	Every strategy

Because log returns add, $R_{T,h} = R_{T,1} + D_{T,h}$, an identity checked in every notebook. **Every strategy in this study enters at the close of the first post-release bar.** This is the single

most important design choice: Section 6.8 shows that an entry at the pre-release price multiplies a strategy's apparent Sharpe by two.

5.2 The event-study regressions (Notebooks 02, 02b)

For each release family and horizon:

$$R_{T,h} = \alpha + \sum_k \beta_k z_{k,T} + \varepsilon_T,$$

with the Notebook 02 components (CPI: headline and core; NFP: payrolls, unemployment rate, average hourly earnings; PCE: headline and core) and **HC3** heteroskedasticity-robust standard errors.

5.3 State interaction tests (Notebooks 03, 03b)

$$r_T = a + b x_T + c s_T + d (x_T \times s_T) + \varepsilon_T,$$

where s_T is a state percentile in $[0, 1]$. Because x is signed so that $b > 0$, a positive d means a **stronger reaction in the high state**, and d is the change in the surprise beta from the lowest to the highest state. Each interaction is tested four ways, all **pre-registered**:

1. HC3 t-test and a **2,000-draw permutation test** (shuffling the state across events);
2. **Benjamini-Hochberg FDR** correction across the 48 primary tests ($q < 0.10$);
3. **Subperiod stability**: the same sign in 2016-2020 and 2021-2026;
4. **Horizon consistency**: the same sign across most of the 9 horizons.

A (state, family) pair is carried into Notebook 04 only if it passes all four.

5.4 Walk-forward backtest (Notebooks 04-08)

- **Estimation.** Every coefficient, sign, screen and gate is estimated using only releases **before** the one being traded. The first 36 releases of a family are training only.
- **Evaluation.** 2019-01 to 2026-06, the same for every notebook and market.
- **Positions.** One contract per trade (or a fraction, for the volatility-sized books), held for 30 minutes.
- **Costs.** Base case: **2 ticks of slippage (1 per side) + \$4.50 commission per round trip**, converted to bps with the pre-release price:

$$\text{cost}_{\text{bps}} = \frac{2 \times \text{tick} + 4.50/\text{point value}}{P_{T-1}} \times 10^4.$$

Sensitivities run from 0 to 8 ticks, and Notebook 08 also uses the peer assumption of $\frac{1}{4}$ tick per side.

- **Sharpe ratio.** Monthly net returns (months without trades count as zero) $\times \sqrt{12}$. The peer-format tables use daily returns $\times \sqrt{252}$, as on the team's slides.

5.5 Statistical safeguards

Risk	Safeguard
Look-ahead bias	Causal z-scores, trailing percentiles, walk-forward estimation, explicit checks (asof < T, pre-close matches)
Multiple testing	Pre-registered states and targets, BH-FDR, permutation tests
Luck in the direction	Random-sign test: 1,000 books with the same trades and costs but random directions; p = share with a Sharpe at least as high

Risk	Safeguard
Uncertain Sharpe	Moving-block bootstrap (3-month blocks) of the monthly Sharpe; paired bootstrap of Sharpe differences
Regime dependence	Sharpe excluding 2022; 2019–2021 vs 2022–2026
Execution	Cost sensitivity; entry delays of 1, 2 and 5 minutes; holding periods of 5–60 minutes
Code errors	Cross-notebook equality checks (every notebook reproduces the previous one’s trade returns to $< 10^{-13}$ bps); DTW validation on synthetic data

5.6 Release universe (Notebooks 05–07)

To trade more than CPI, an **impact screen** (Jack Duncan’s method) ranks release families each year using only history:

$$\text{impact}_f = \frac{\text{mean } |30\text{-min reaction}| \text{ on family } f\text{'s releases}}{\text{mean } |30\text{-min reaction}| \text{ at the same clock time on other families' releases}}.$$

The **top 10 families** each year are traded (CPI is always included). This yields about 108 release timestamps a year.

5.7 Software

All analysis is in Python (numpy, pandas, matplotlib). HC3 regression, BH-FDR, the permutation and random-sign tests, the block bootstrap, and a vectorised DTW with a Sakoe-Chiba band are implemented in `src/` without external statistics libraries.

Module	Content
<code>src/state_conditioning.py</code>	Loading and roll-adjusting minute bars, daily closes, state features, HC3 OLS, interaction and magnitude tests, BH-FDR
<code>src/walk_forward.py</code>	Strategy specifications, walk-forward engine, costs, performance metrics, bootstrap, peer-format daily metrics
<code>src/surprise_universe.py</code>	Bloomberg loader, release families, causal surprises, price measures, impact screen, line signs, positions
<code>src/dtw_signal.py</code>	Price paths, vectorised DTW, walk-forward DTW signal, rank gate, five-window ladder
<code>src/multi_asset.py</code>	ES/NQ/ZN contract specs, the full Notebook 05+06 pipeline per market, portfolio weights
<code>src/team_replication.py</code>	Rebuild of Jack Duncan’s DTW ladder from its published specification (Notebook 08)

6. Results, notebook by notebook

Each subsection follows the same structure: **question** → **design** → **results** → **interpretation**. ES results come first; NQ and ZN (the “b” notebooks, which run identical code on the other two markets) follow.

6.1 Notebook 01: building causal macro surprises

Question. Can the Bloomberg release workbook be turned into a clean, comparable, look-ahead-free surprise for every release?

Design. The 17 yearly sheets are stacked into one table of (ticker, release time, event name, survey median, actual). Release times are converted from ET to UTC. For the core event study, one row is built per release of CPI, the Employment Report (NFP) and PCE, with a standardised surprise for

each component (headline CPI, core CPI; payrolls, unemployment rate, average hourly earnings; headline PCE, core PCE).

Result. A panel of **363 releases** (2016–2026) with standardised surprises for seven components, saved as `data/processed/macro_event_model_data_2016_2026.parquet` and used by every later notebook. Notebook 05 extends the same logic to all 179 lines (17,328 causal surprises, 75 release families).

6.2 Notebooks 02 and 02b: which releases move which market?

Question (RQ1). Which releases move the market, how much, and how much of the move is explained by the surprise?

Design. Event study of the 363 CPI, NFP and PCE releases: the reaction from the last pre-release price to 1, 5, 15, 30 and 60 minutes after the release, regressed on the surprise components with HC3 errors. Notebook 02 covers ES; Notebook 02b repeats it for NQ and ZN with ES as the reference, and adds the **tradeable drift** from T+1.

6.2.1 ES (Notebook 02)

Release	Mean reaction at 30 min (bps)	R ² at 1 min	R ² at 5 min	R ² at 30 min	Headline beta, 5 min (bps per 1 SD)
CPI	38.5	0.30	0.27	0.25	−20.8 (t = −3.1)
NFP	29.6	0.03	0.03	0.04	+2.8 (payrolls, not significant)
PCE	14.7	0.14	0.10	0.03	−3.8

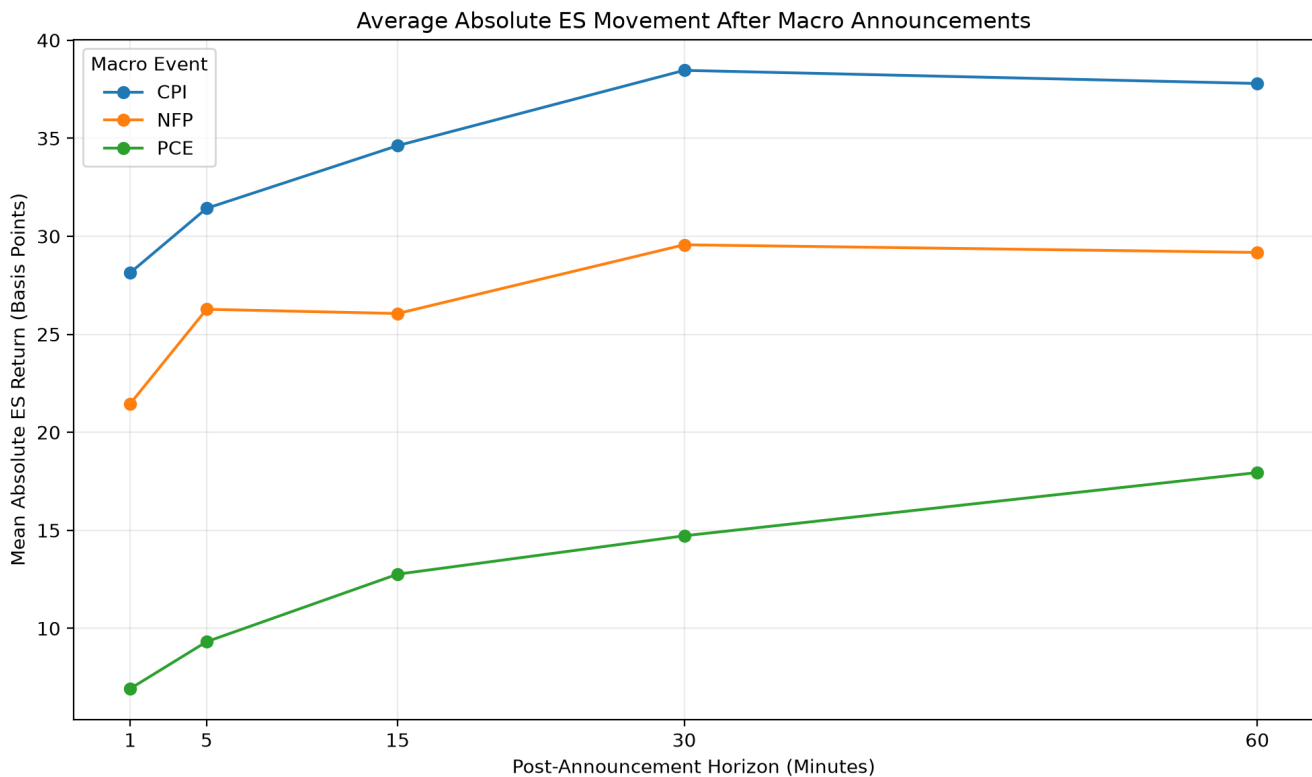


Figure 3. Mean absolute ES reaction by release and horizon.

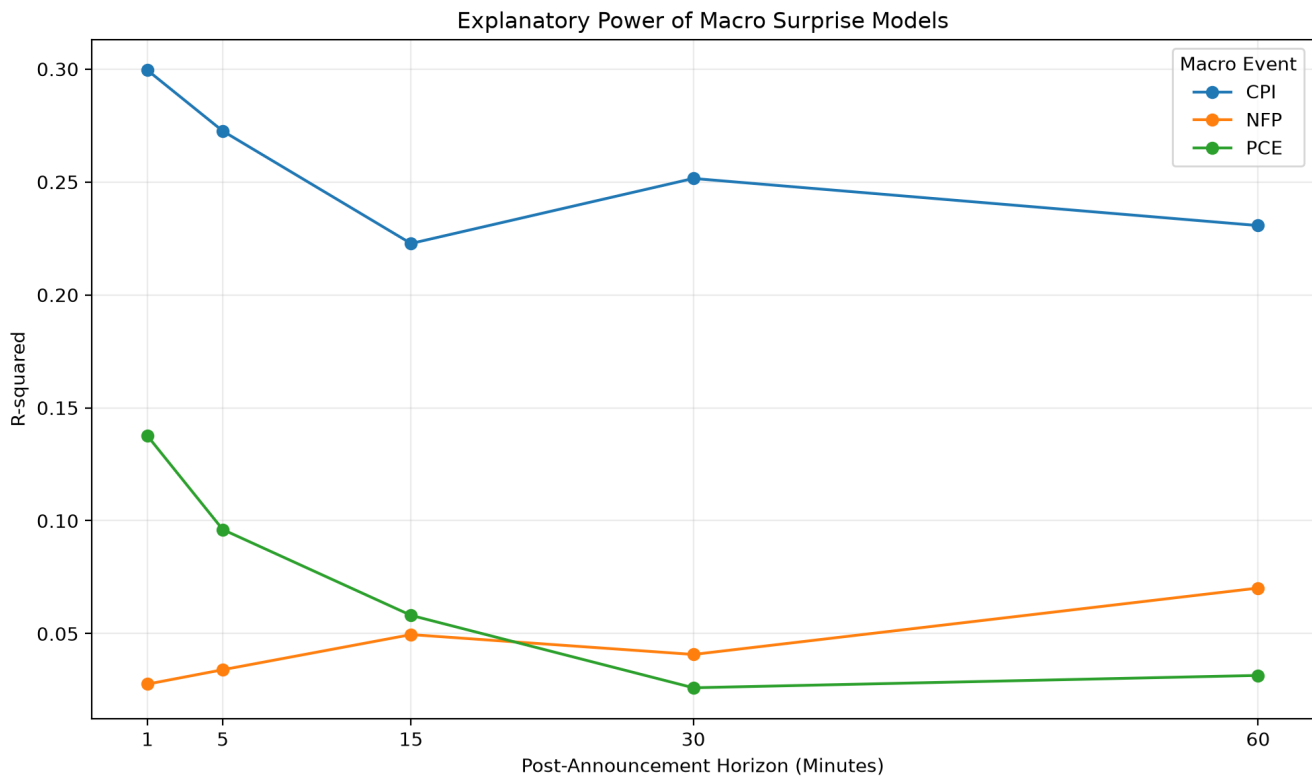


Figure 4. How much of the ES reaction the surprise explains (R^2), by release and horizon.

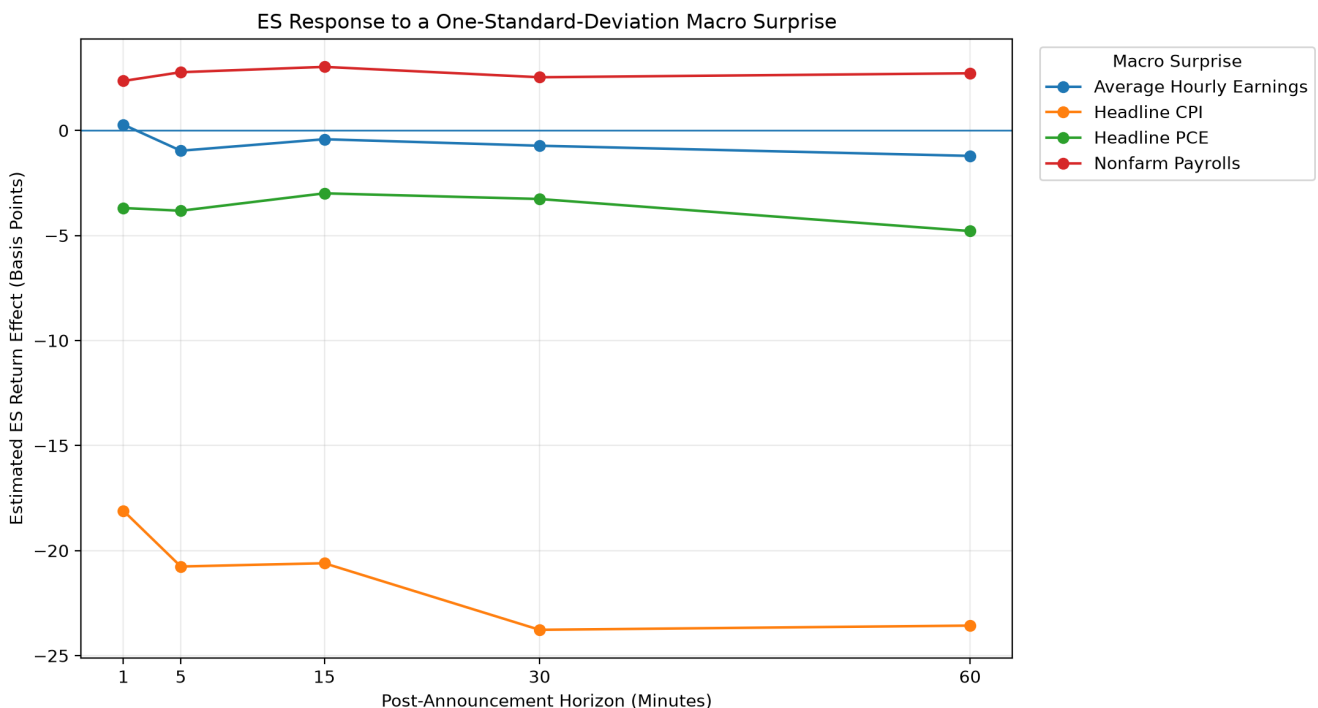


Figure 5. Surprise coefficients (bps per 1-SD surprise) and their significance.

The headline CPI coefficient is **-18 to -24 bps per 1-SD surprise** at every horizon from 1 to 60 minutes, all significant at 1%. NFP moves ES almost as much as CPI (30 bps on average) but in a direction the surprise hardly explains (R^2 3-7%), because payrolls, unemployment and wages often point in different directions and the market weighs growth against rates. The only other significant effect at 5% is average hourly earnings at 60 minutes (-1.2 bps).

Release	Main signal	Typical ES response	Impact rank	Directional rank	R ² range	Strength
CPI	Headline CPI	Hotter inflation → ES lower	1	1	22-30%	Strong
NFP	Payrolls / wages	Strong payrolls → modestly higher; hot wages → lower	2	3	3-7%	Weak / mixed
PCE	Headline PCE	Hotter inflation → ES lower	3	2	3-14%	Moderate at short horizons

6.2.2 NQ and ZN (Notebook 02b)

Market	Release	Mean 30-min reaction (bps)	Headline beta, 5 min	t	R ² , 5 min	Drift beta T+1→T+30	t
ES	CPI	38.5	-20.8	-2.8	0.27	-5.7	-1.4
NQ	CPI	53.6	-27.7	-2.7	0.30	-4.6	-0.9
ZN	CPI	22.5	-9.9	-2.8	0.30	-0.06	-0.04
ES	NFP	29.6	+2.8	1.0	0.03	+0.2	0.1
NQ	NFP	36.0	+2.5	1.0	0.01	-0.6	-0.4
ZN	NFP	23.6	-1.5	-0.8	0.03	+0.1	0.1
ES	PCE	14.9	-3.8	-1.8	0.10	+0.4	0.2
NQ	PCE	18.5	-5.2	-2.0	0.10	+1.3	0.5
ZN	PCE	8.3	-0.9	-1.0	0.04	+0.5	0.5

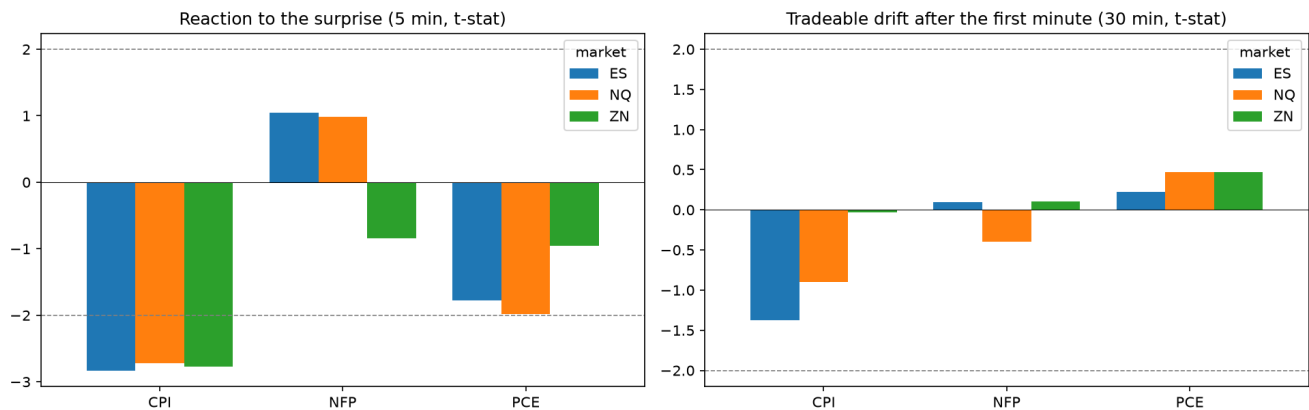


Figure 6. Reaction (from the pre-release price) vs tradeable drift (from T+1) for the three markets.

Interpretation.

- **CPI is the valuable release in all three markets.** It explains 27-30% of the five-minute reaction everywhere, with the economically expected sign: a hot print lowers equities and bond prices.
- **NQ reacts about 35% more than ES** (-27.7 vs -20.8 bps per SD), consistent with the higher rate sensitivity of long-duration technology stocks.
- **ZN reacts in bps about half as much as ES**, but a ZN bp is a larger move in yield terms; its R² is as high as ES's.
- **The drift is the problem.** Measured from T+1 with the multivariate specification, the 30-minute drift is -5.7 bps per SD in ES, -4.6 in NQ and essentially **zero in ZN**. None is individually significant with 120 CPI releases, which is why the trading tests in Notebooks 04-05 (which aggregate many releases and use walk-forward signs) are the decisive evidence.

6.3 Notebooks 03 and 03b: does the pre-event state matter?

Question (RQ2, RQ3). Does the same surprise have a different effect depending on the market's state before the release? And how much of the reaction is tradeable?

Design. Six pre-registered states (Section 3.3), two pre-registered targets (the 5-minute reaction `ret_5m` and the 30-minute tradeable drift `drift_30m`), bucket betas by tercile, interaction regressions, permutation tests, BH-FDR, subperiod stability, magnitude regressions, and a four-criterion selection rule (Section 5.3).

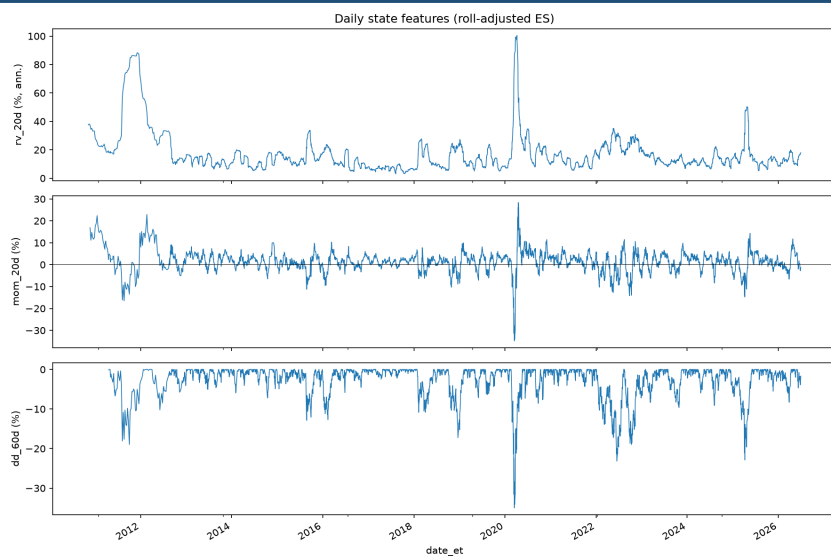
6.3.1 The surprise signal and the tradeable part

	ES	NQ	ZN
corr(x, 5-min return), CPI	0.51	0.54	0.53
corr(x, 5-min return), PCE	0.31	0.32	0.19
corr(x, 5-min return), NFP	0.13	0.03	0.16 (sign reversed for bonds)
CPI: mean 30-min drift (bps)	~20	26.3	10.5
CPI: 5-min reaction per 1-SD surprise (bps)	~35	36.9	14.2
CPI: 30-min drift per 1-SD surprise (bps)	~9	8.5	0.8

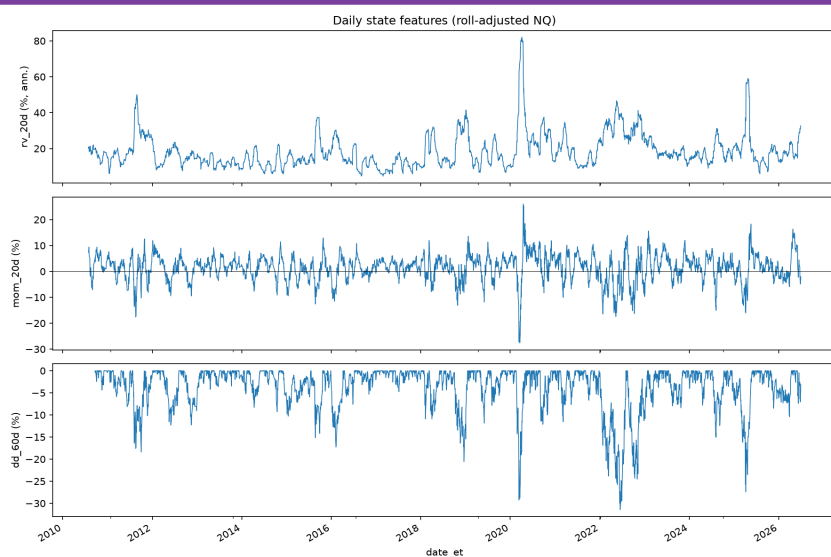
The surprise signal is correctly signed in every market. In ES and NQ roughly a quarter of the CPI reaction remains as a drift after the first minute; in ZN almost none does. This one line explains most of the trading results that follow.

6.3.2 The market's daily state

ES | E-mini S&P 500



NQ | E-mini Nasdaq-100



ZN | 10-year T-note

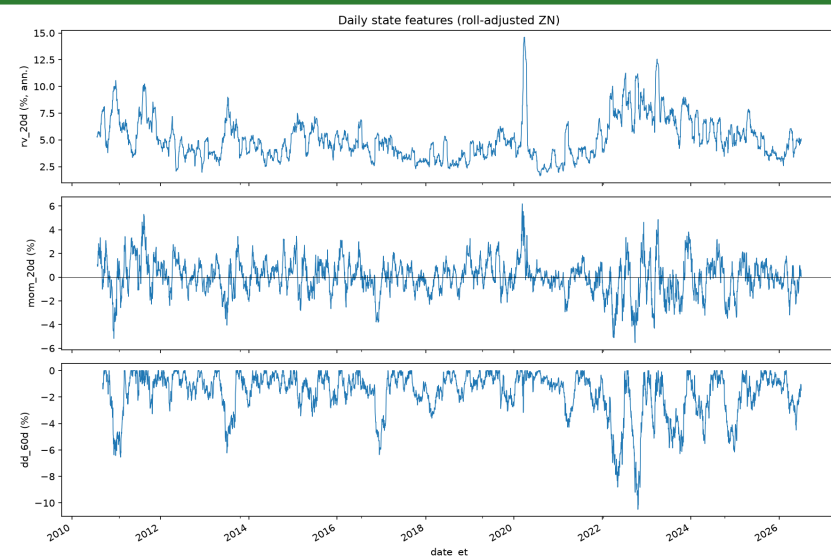


Figure 7. Daily state features of ES (top), NQ (middle) and ZN (bottom), 2010–2026: 20-day realised volatility (annualised %), 20-day momentum (%) and distance from the 60-day high (%). All three are computed from roll-adjusted daily closes.

Reading Figure 7 across markets. The two equity indices share the same history of stress. Volatility spikes in 2011, late 2015, 2018, **March 2020** (the largest: about 100% annualised in ES and about 80% in NQ), through 2022 (NQ stays above 30–40% for most of the year) and again in **April 2025**, and every spike coincides with a deep drawdown and a sharp fall in momentum. NQ’s levels are higher than ES’s throughout (median volatility about 16% vs 13%), and its 2022 drawdown (about –30%) is much deeper than ES’s, because long-duration technology stocks were hit hardest by rising rates. ZN lives on a different scale: its volatility is usually 3–8% and peaks at about 15% in March 2020, and its drawdowns come from **rising yields** (2013 taper tantrum, late 2016, and the 2022 tightening, when it reached –10.5%), not from equity crashes. In March 2020 ZN’s momentum turned sharply positive while equities collapsed: bonds rallied as a safe haven. This is why the equity states behave as one “stress regime” and the bond states do not.

Daily state	ES	NQ	ZN
20-day realised vol, median	~13%	~16%	~4.7%
20-day realised vol, maximum	~100% (March 2020)	~82%	~15%
Deepest 60-day drawdown	–35%	–31%	–10.5%
Spearman corr(vol, drawdown) at events	–0.6 to –0.8	–0.67 to –0.75	–0.15 to –0.21

In the equity markets volatility, momentum and drawdown move together as one “stress regime”; in ZN they are largely separate.

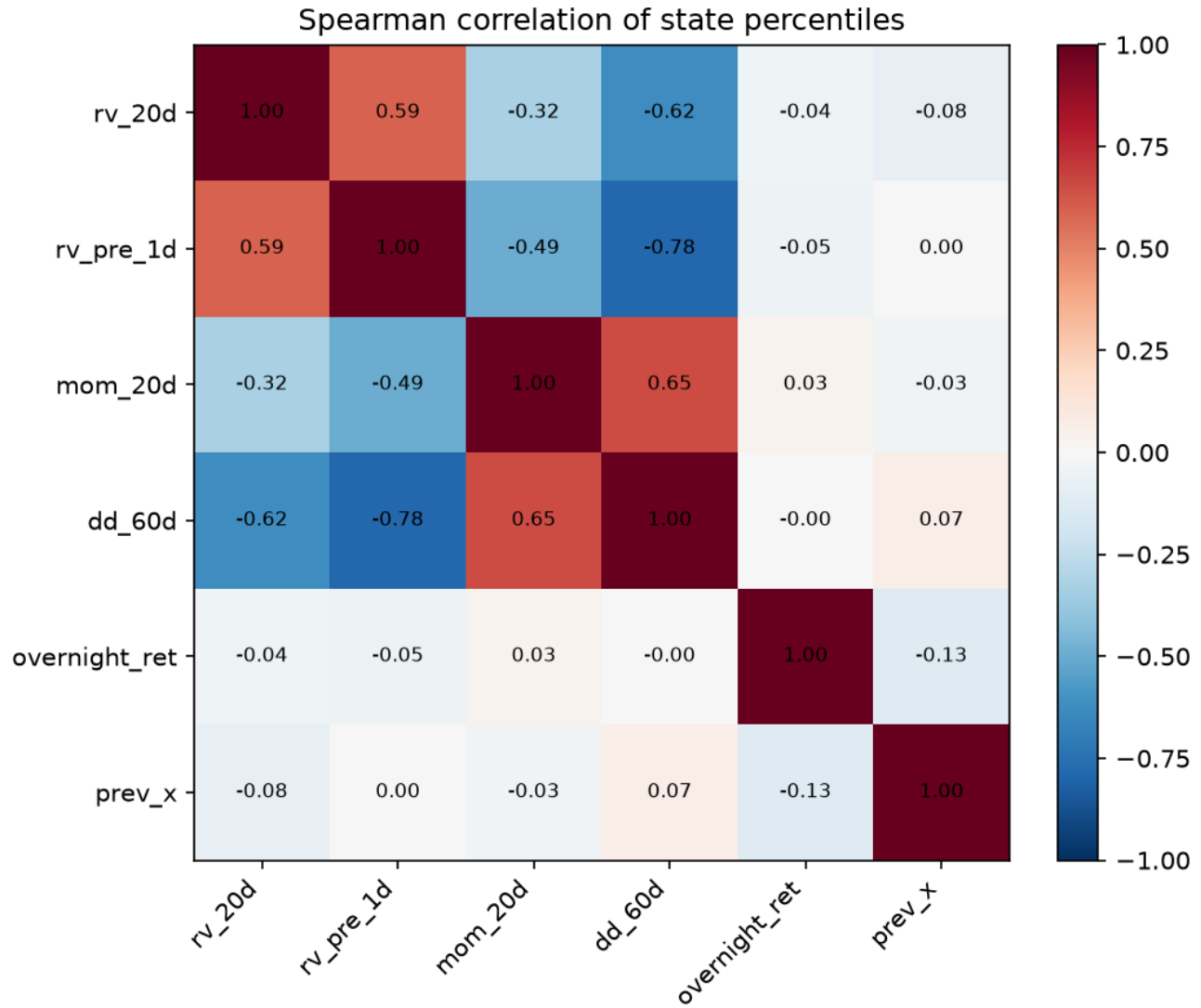


Figure 8. Spearman correlation between the six pre-registered state percentiles (ES).

6.3.3 Bucket betas: CPI response by state tercile

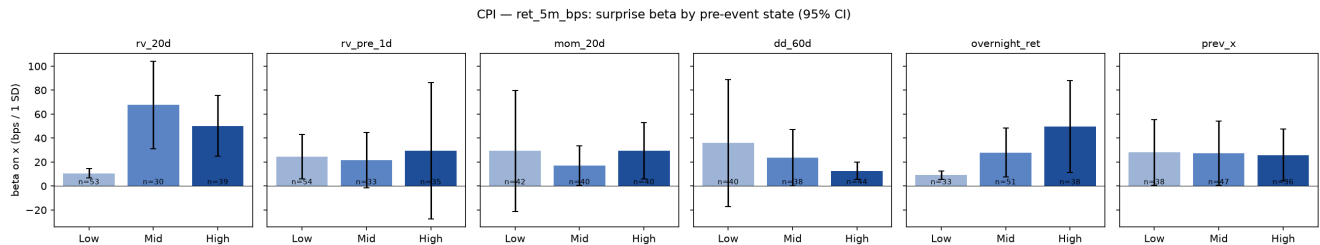


Figure 9. ES: CPI surprise beta on the 5-minute reaction, by tercile of each state.

CPI beta on the 5-min reaction (bps per 1-SD good-news surprise)	Low	Mid	High
ES · 20-day volatility	11	50-68	50-68
NQ · 20-day volatility	18	70	58
ZN · 20-day volatility	15	12	19
ES · overnight return	9	28	50
NQ · overnight return	16	28	76

CPI beta on the 5-min reaction (bps per 1-SD good-news surprise)	Low	Mid	High
ZN · overnight return	9	14	18

In **ES** and **NQ**, **CPI hits much harder when volatility is elevated**, and the response rises monotonically with the overnight move. In **ZN** there is no volatility pattern; only the overnight pattern repeats.

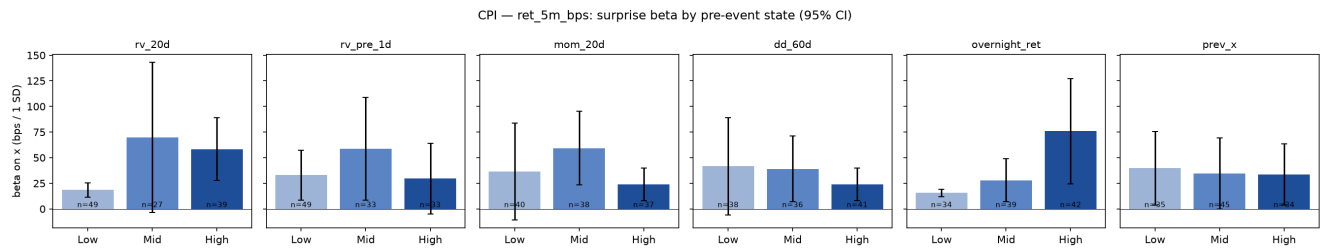


Figure 10. NQ: CPI surprise beta by state tercile.

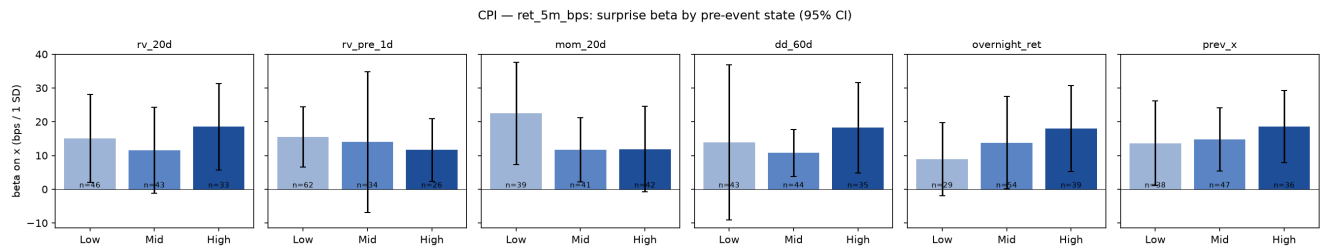
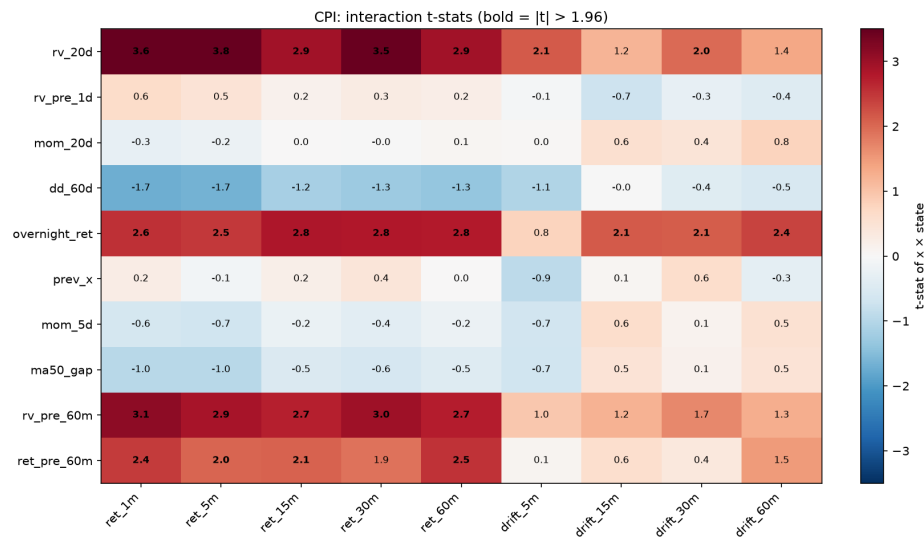


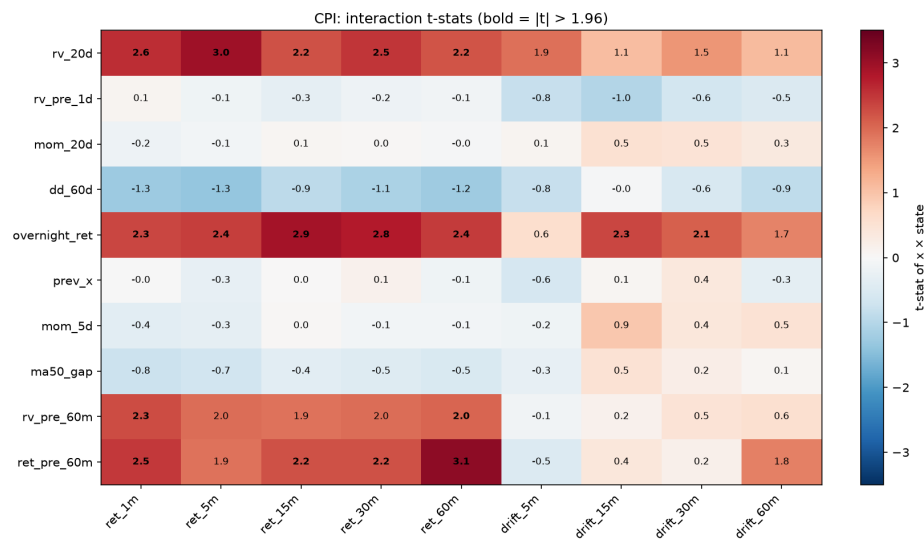
Figure 11. ZN: CPI surprise beta by state tercile.

6.3.4 Interaction tests, multiple testing and stability

ES | E-mini S&P 500



NQ | E-mini Nasdaq-100



ZN | 10-year T-note

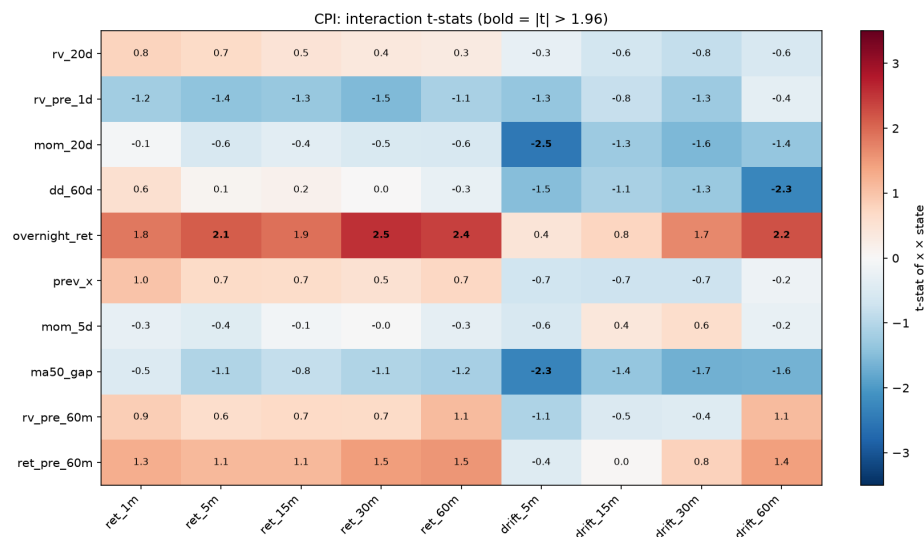


Figure 12. t -statistics of the $\text{CPI} \times \text{state}$ interaction d for ES (top), NQ (middle) and ZN (bottom), across the ten states (rows) and nine targets (columns: the 1–60 minute reactions and the 5–60 minute drifts). Red: the state strengthens the CPI response; blue: it weakens it. A credible effect is a row of same-coloured cells; a single bold cell among about 90 is what chance produces.

Reading Figure 12 across markets. In **ES** the rv_20d row is dark red on every reaction horizon ($t = 2.9\text{--}3.8$ from 1 to 60 minutes) but much lighter on the drift columns ($t = 1.2\text{--}2.1$): volatility mainly amplifies the **first-minute jump**. The overnight_ret row is red on both reactions and drifts ($t \approx 2.1\text{--}2.8$), and the exploratory 60-minute pre-release volatility and return rows repeat the same message. **NQ** shows the same red rows, slightly weaker (rv_20d $t = 2.2\text{--}3.0$ on the reactions, $1.1\text{--}1.9$ on the drifts; overnight_ret up to 2.9), which is strong evidence that the ES pattern is not a fluke of one contract. **ZN** looks different: the rv_20d row is pale ($|t| < 1$), overnight_ret is the only red row ($t \approx 1.8\text{--}2.5$), and the drift columns are mostly blue for momentum, distance from the 50-day average and drawdown: in bonds, a strong recent trend weakens the post-release drift rather than strengthening it. The significant ZN state effect (rv_pre_1d) appears only when all releases are pooled; for CPI alone its row is mildly blue (t between -0.4 and -1.5).

	ES	NQ	ZN
Strongest primary interaction	CPI $\times$ rv_20d on ret_5m , $t = \mathbf{3.8}$ (+61 bps from calmest to most volatile)	CPI $\times$ rv_20d on ret_5m , $t = \mathbf{3.0}$ (+58 bps)	$\text{rv_pre_1d} \times$ all releases on drift_30m , $t = \mathbf{-3.2}$ (–4.6 bps)
Other notable	NFP $\times$ vol, $t = -3.0$	CPI $\times$ overnight, $t = 2.4$	CPI $\times$ overnight, $t = 2.1$
Tests significant at 5% (expected by chance: 2.4 of 48)	7	3 (4 by permutation)	3 (3 by permutation)
Survive BH-FDR ($q < 0.10$)	2	0	1
Stable in both subperiods?	CPI $\times$ vol: same sign, significant only after 2021	Same as ES	Yes: $t = -2.9$ before 2021, -2.2 after
Selected for Notebook 04	$\text{rv_20d} \times \text{CPI}$, $\text{rv_20d} \times \text{NFP}$	none	$\text{rv_pre_1d} \times$ all releases

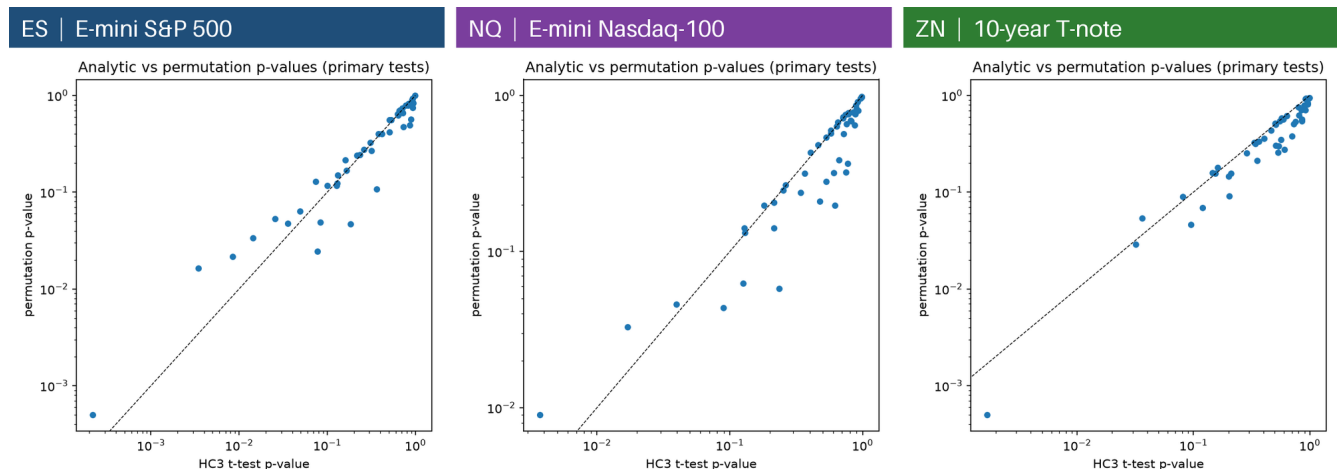


Figure 13. Analytic (HC3 t -test) p -values against permutation p -values (2,000 shuffles of the state across events) for the 48 primary interaction tests, on log scales, for ES (left), NQ (centre) and ZN (right). The dashed line is the 45° line; points in the lower-left corner are the significant tests.

Reading Figure 13 across markets. In all three markets the points lie close to the 45° line, so the analytic HC3 tests agree with a test that makes no distributional assumption: fat tails and outlying releases do not drive the results. What differs is how many points reach the lower-left corner: **7 below 5% in ES** (2.4 expected by chance), with one test at $p \approx 0.0005$ in both methods; **3–4 in NQ**, the strongest at about 0.004–0.009; and **3 in ZN**, with one very strong test ($p \approx 0.002$ analytic, 0.0005 permutation). Only ES has clearly more significant results than chance, which is why two ES tests survive the FDR correction, none in NQ and one in ZN (the pooled rv_pre_1d effect).

6.3.5 The state predicts the size of the move

Magnitude tests surviving FDR (of 360)	ES	NQ	ZN
Count	198	154	0
Example	High rv_{20d} : absolute move ~20 bps larger ($t \approx 4-5$)	High rv_{20d} : 30-min drift ~18 bps larger ($t = 4.2$); deep drawdown ~23 bps larger ($t = -4.8$)	Best: previous surprise $\times$ PCE, $q = 0.10$

Interpretation of Section 6.3.

- The pre-event state **matters for the size** of equity-index moves: in stressed markets everything moves more. This is the most robust result of the notebook in ES and NQ, and it is useful for **risk sizing**, not direction.
- The state **also changes the CPI surprise beta** in ES and NQ (about +60 bps from calm to volatile markets), but mainly on the **untradeable 5-minute reaction**, mainly **after 2021**, and in NQ not strongly enough to survive multiple-testing correction.
- In ZN the only robust state effect is **negative and small**: after a volatile day, the post-release drift follows the surprise less. It is the only state effect in the whole study that is significant in both subperiods, but at -4.6 bps across the full range it is economically too small to trade.

6.4 Notebooks 04 and 04b: can the reaction be traded in real time?

Question (RQ4). Could a trader have made money from CPI surprises in real time, after costs, using only information available at each release? And does conditioning on the pre-event state improve the strategy out of sample?

Design. Entry at the close of the first post-release bar, exit 30 minutes after the release. Five strategies (Section 3.4: naive sign, surprise only, surprise + state, state filter, adaptive state), re-estimated walk-forward. Universes: CPI; CPI + PCE; CPI + PCE + NFP. Evaluation 2019-2026, costs 2 ticks + \$4.50. The state models use rv_{20d} (the ES selection) in all three markets.

6.4.1 Main result: CPI, net of costs

CPI, 2019-2026	ES	NQ	ZN
Average round-trip cost	1.7 bps	0.7 bps	3.0 bps
Naive sign: trades / net bps per trade / Sharpe	86 / 9.8 / 1.03	79 / 13.6 / 1.00	86 / -0.3 / -0.05 (gross 0.53)
Surprise only: trades / net bps / Sharpe	52 / 12.8 / 0.99	62 / 12.1 / 0.79	2 / -3.2 / -0.51
Surprise + State	51 / 10.4 / 0.77	64 / 11.2 / 0.75	5 / -3.6 / -0.40
State filter	28 / 23.0 / 1.07	36 / 19.9 / 0.82	1 / -3.9 / -0.37
Adaptive state	49 / 12.7 / 0.95	64 / 11.2 / 0.75	2 / -3.2 / -0.51
Surprise only: 90% bootstrap CI of Sharpe	0.25 to 1.72	0.12 to 1.47	—
Surprise only: hit rate / t-stat of average trade	64% / 2.8	58% / 2.2	—

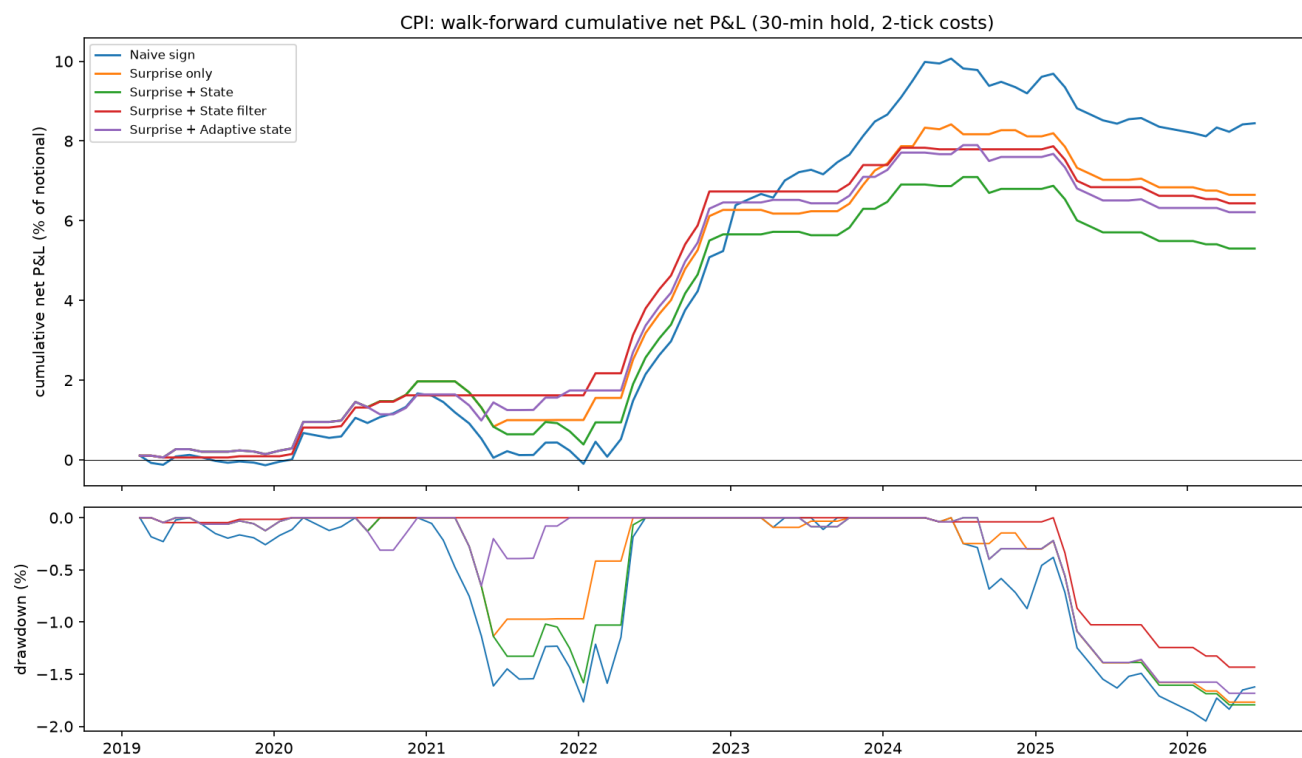


Figure 14. ES: walk-forward cumulative net P&L of the CPI strategies and drawdowns.

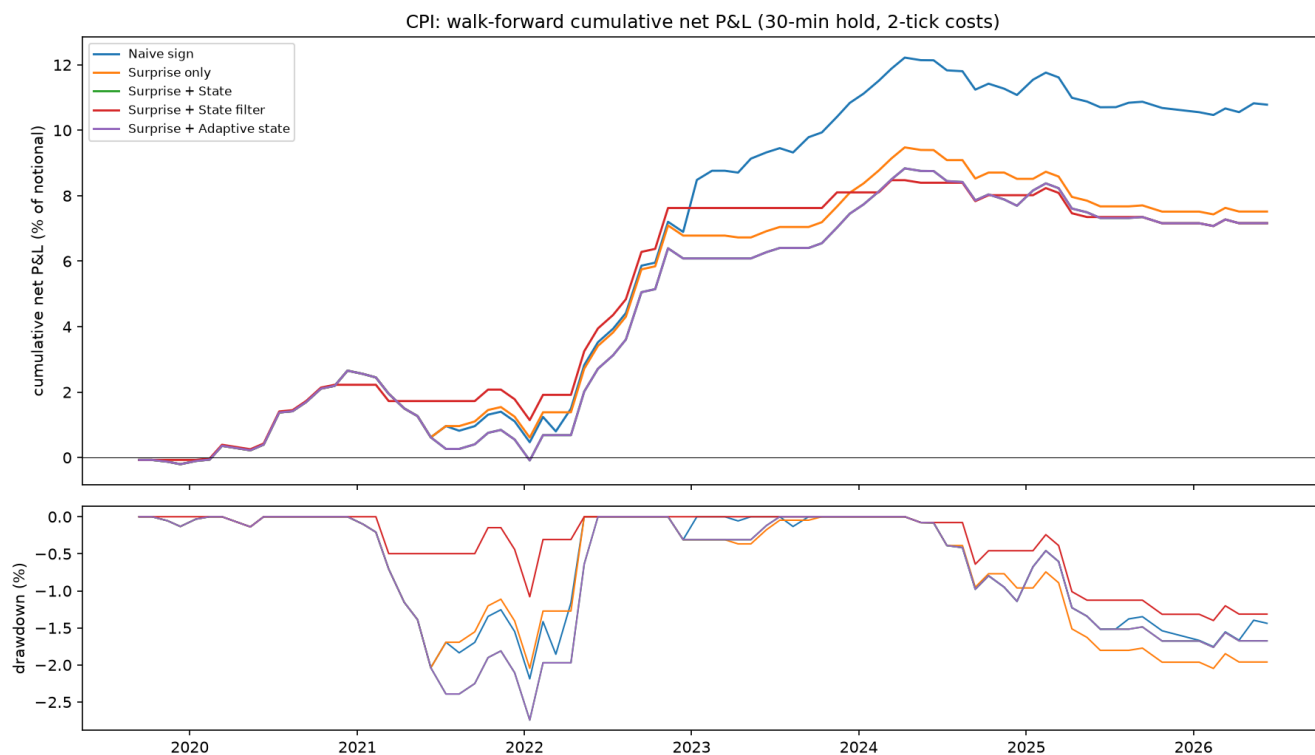


Figure 15. NQ: the same test.

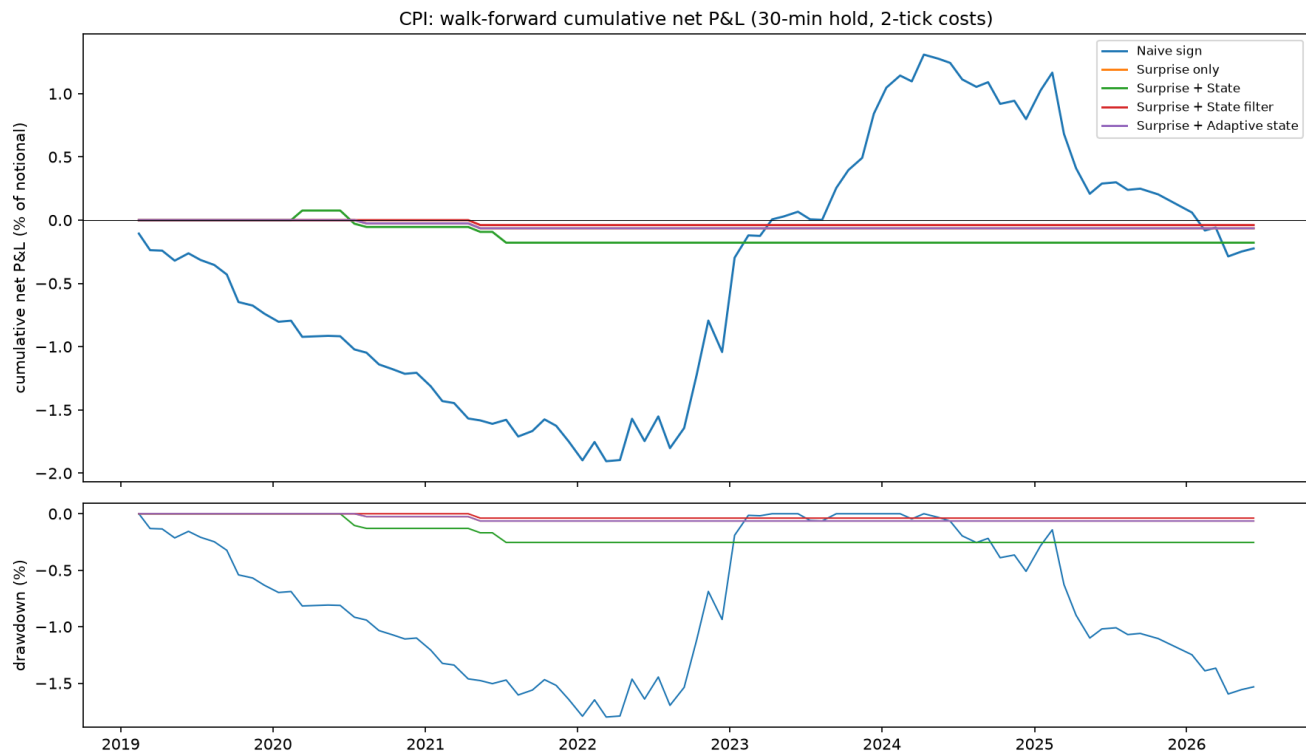


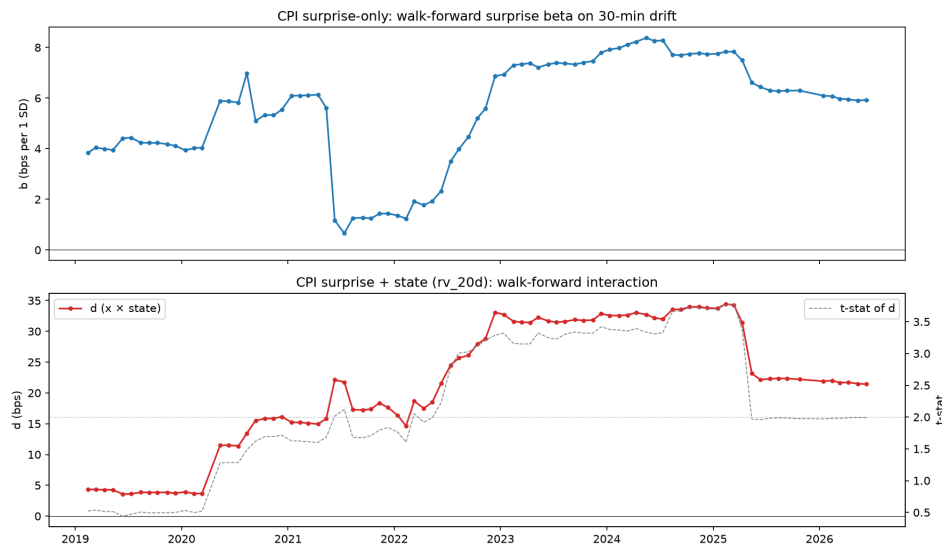
Figure 16. ZN: the same test. The fitted models almost never trade because the predicted move rarely exceeds the cost.

Interpretation.

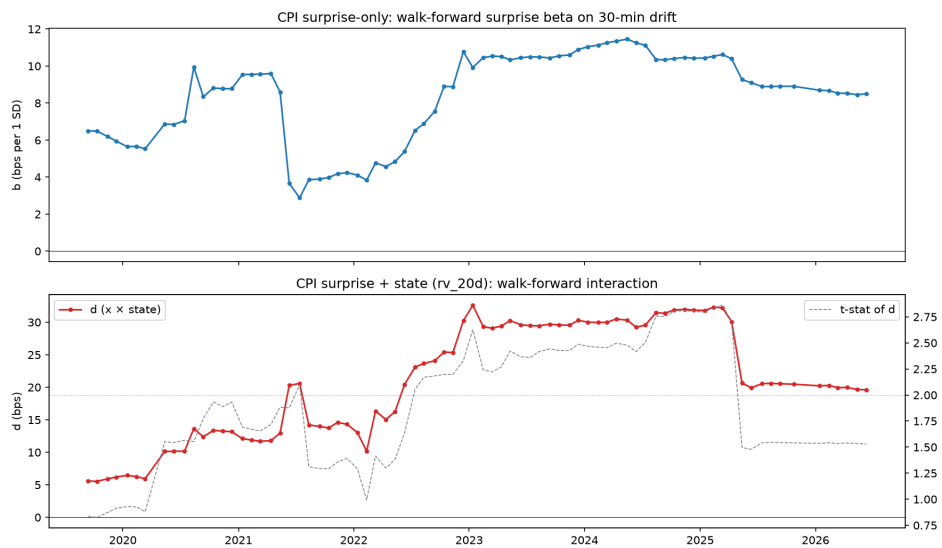
- **ES and NQ: yes, CPI surprises are tradeable.** A net Sharpe of about 1.0 in ES and 0.8–1.0 in NQ, statistically positive, with 8–11 trades a year.
- **The naive sign rule is as good as or better than the fitted model.** In ES the two are equal; in NQ the naive rule is *significantly* better (Sharpe +0.21, $P(\text{not better}) = 0.04$), because with costs below 1 bps it pays to trade every CPI release rather than wait for a large predicted move. The edge comes from the **economic direction** of the surprise, not from estimating a model.
- **ZN: no.** The naive CPI rule makes +2.7 bps per trade before costs (gross Sharpe 0.53), about the same as the 3 bps cost. The fitted models correctly refuse to trade.

6.4.2 When could a trader have known about the state effect?

ES | E-mini S&P 500



NQ | E-mini Nasdaq-100



ZN | 10-year T-note

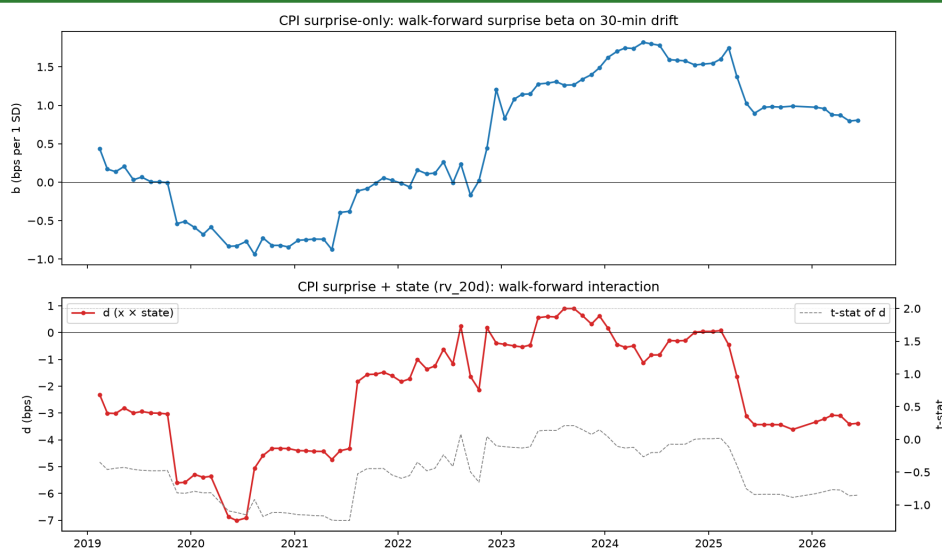


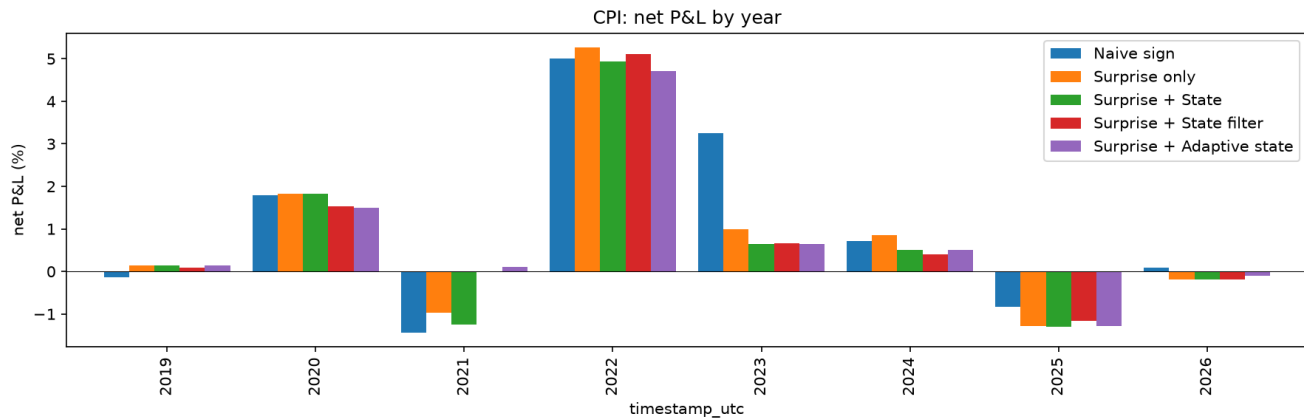
Figure 17. Walk-forward estimates at each CPI release of the surprise beta $\hat{b}_t$ (surprise-only model and surprise + state model) and the state interaction $\hat{d}_t$ with its t-statistic, for ES (top), NQ (middle) and ZN (bottom). Each point uses only earlier releases.

Reading Figure 17 across markets. In **ES** the CPI beta on the 30-minute drift starts at about 4 bps per 1-SD in 2019, **collapses to about 1 bp in mid-2021** (when inflation surprises first appeared and the market's reaction was confused), then climbs steadily through 2022 to about **8 bps in 2023-2024** before easing to 6 bps. The interaction $\hat{d}_t$ rises from about 4 bps ($t \approx 0.5$) in 2019 to about 33 bps ($t \approx 3.5$) from 2023, and its t-statistic first crosses 2 only in mid-2022; after a sharp drop in 2025 it stays at about $t = 2$. **NQ** follows exactly the same shape at a higher level (beta 6 $\rightarrow$ 3 in 2021 $\rightarrow$ 10-11 in 2023-2024 $\rightarrow$ 8.5; the interaction's t-statistic peaks at about 2.8 in 2024-2025 and falls to 1.5). In **ZN** the beta stays between -0.9 and $+1.8$ bps, positive only since late 2022, and the interaction is small and negative, with t between -1.2 and $+0.2$ throughout. For a trader in real time, the state effect was invisible until 2022 in both equity markets and never existed in bonds.

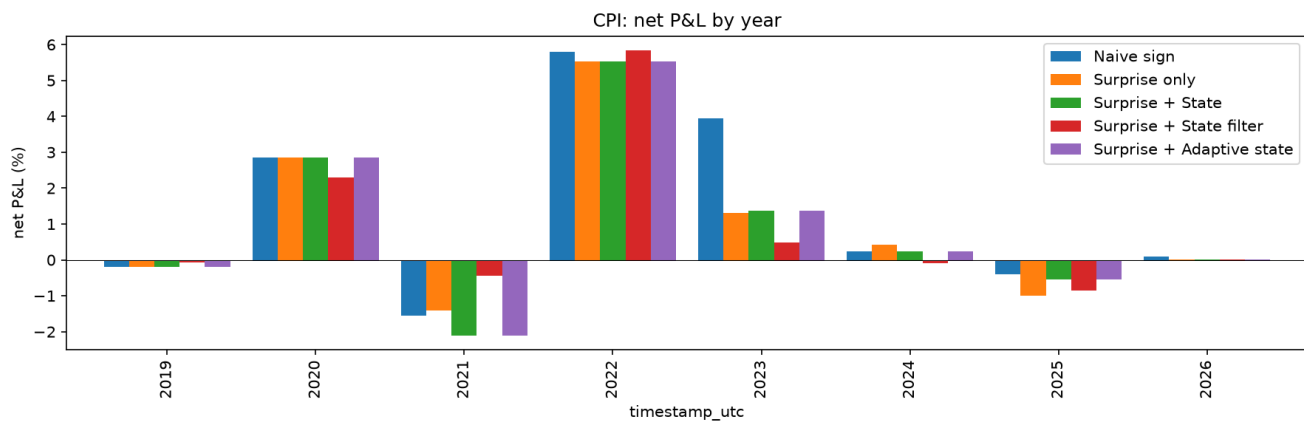
The $\text{CPI} \times \text{rv_20d}$ interaction was **not detectable in real time before 2022**: its training t-stat was 0.5 in 2019 and 1.7-1.8 in 2020-2021, and reached 3.3 only in 2022 (ES). NQ shows the same pattern (0.9 $\rightarrow$ 1.4-1.9 $\rightarrow$ 2.3-2.8 in 2022-2024 $\rightarrow$ 1.5 in 2025-2026). The effect that Notebook 03 finds on the full sample could not have been used by a trader in 2019-2021.

6.4.3 Where do profits come from?

ES E-mini S&P 500



NQ E-mini Nasdaq-100



ZN 10-year T-note

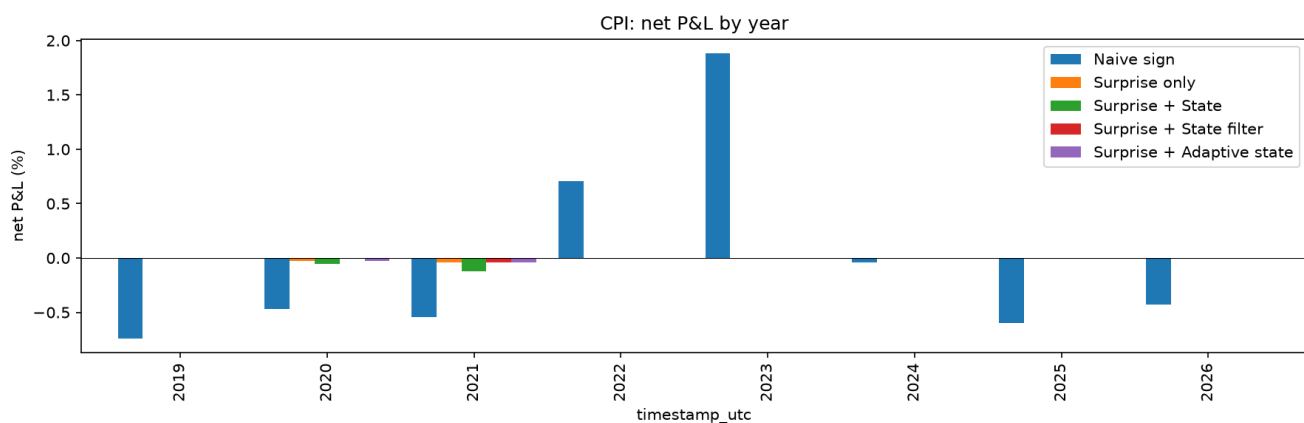


Figure 18. CPI net P&L by calendar year (% of notional) for the five strategies, ES (top), NQ (middle), ZN (bottom).

Reading Figure 18 across markets. ES and NQ tell the same story: small gains in 2020, a loss in 2021 (the first months of the inflation surprise, when the market's reaction function was changing), a **large gain in 2022 (+5 to +6% in every strategy)**, a further gain in 2023 (largest for the naive rule: +3.2% ES, +3.9% NQ), and small losses in 2025. All five strategies move together, so the

differences between models are small compared with the differences between years. In ZN only the naive rule trades, and it earns only in **2022-2023**, the years of the fastest Fed tightening.

CPI surprise-only net P&L (%)	2019	2020	2021	2022	2023	2024	2025	2026	Share from 2022
ES	+0.1	+1.8	-1.0	+5.3	+1.0	+0.9	-1.3	-0.2	79%
NQ	-0.2	+2.9	-1.4	+5.5	+1.3	+0.4	-1.0	0.0	74%

CPI net Sharpe	ES naive	ES surprise	NQ naive	NQ surprise	ZN naive
Full period	1.03	0.99	1.00	0.79	-0.05
Excluding 2022	0.58	0.32	0.68	0.35	-0.27
2019-2021	0.10	0.49	0.37	0.41	-2.60
2023-2026	0.90	0.17	0.91	0.27	0.35

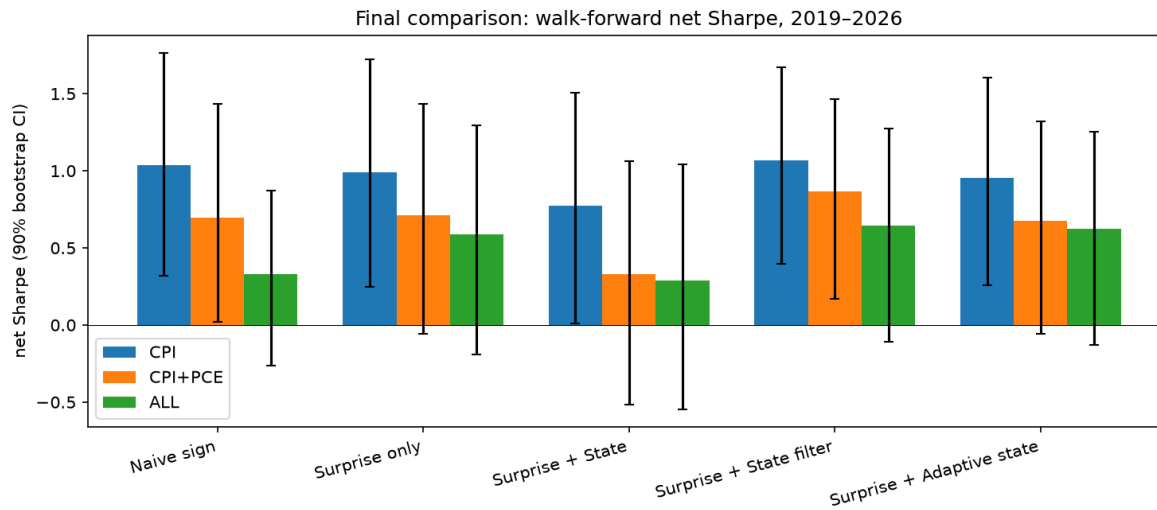
By volatility state, low-volatility CPI trades earn almost nothing (ES +0.9, NQ +1.4 bps per trade), while mid- and high-volatility trades earn +20 to +35 bps. The state information is real, but low-volatility trades are close to zero rather than losers, so skipping them adds little.

6.4.4 Does adding the state beat surprise only?

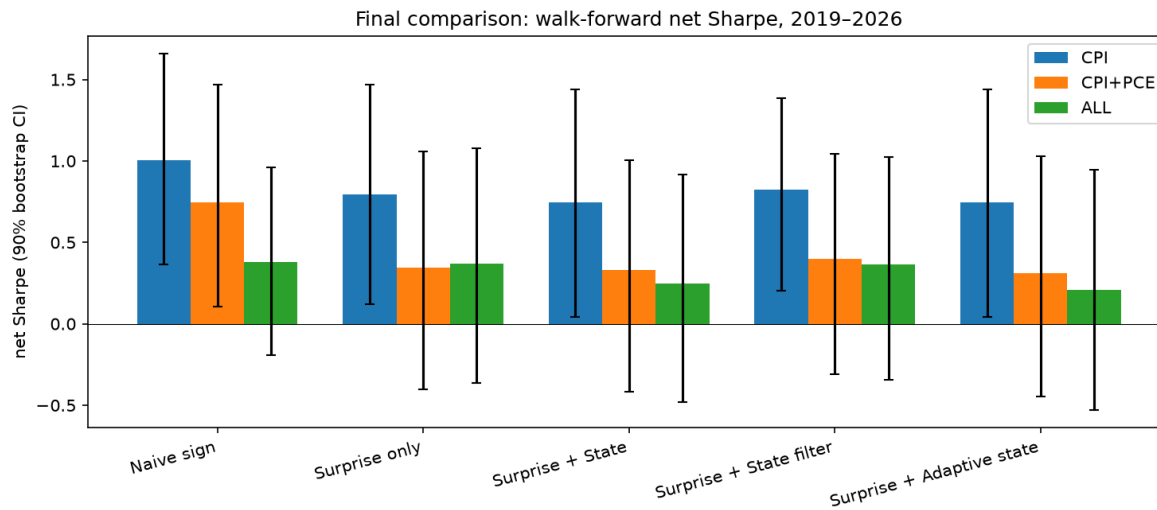
CPI, Sharpe change vs surprise only (P(not better))	ES	NQ
Surprise + State	-0.21 (0.93)	-0.05 (0.68)
State filter	+0.08 (0.40)	+0.03 (0.47)
Adaptive state	-0.04 (0.59)	-0.05 (0.68)
<i>Naive sign (reference)</i>	<i>+0.04</i>	<i>+0.21 (0.04)</i>

No state model significantly beats surprise only in any market. The state filter improves **trade quality** (ES: 71% hit rate, profit factor 5.2, smallest drawdown) but not the Sharpe, and it relies on Notebook 03's full-sample choice. The adaptive model, the cleanest out-of-sample test, chose a different state in different years (none in 2019-2021, *rv_20d* in 2022-2024, *overnight_ret* in 2025-2026, the same sequence in ES and NQ): which state matters is not stable.

ES | E-mini S&P 500



NQ | E-mini Nasdaq-100



ZN | 10-year T-note

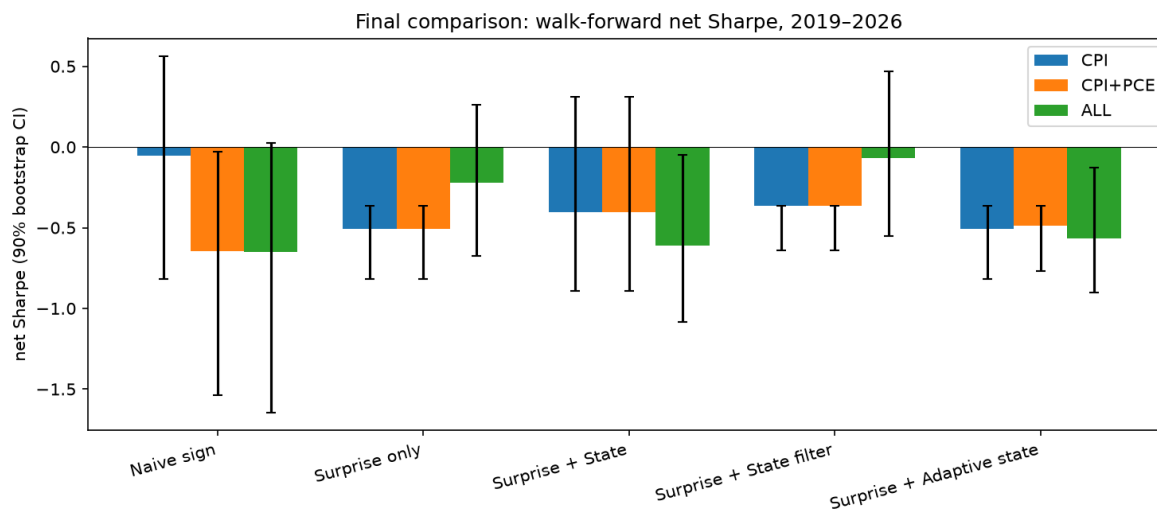
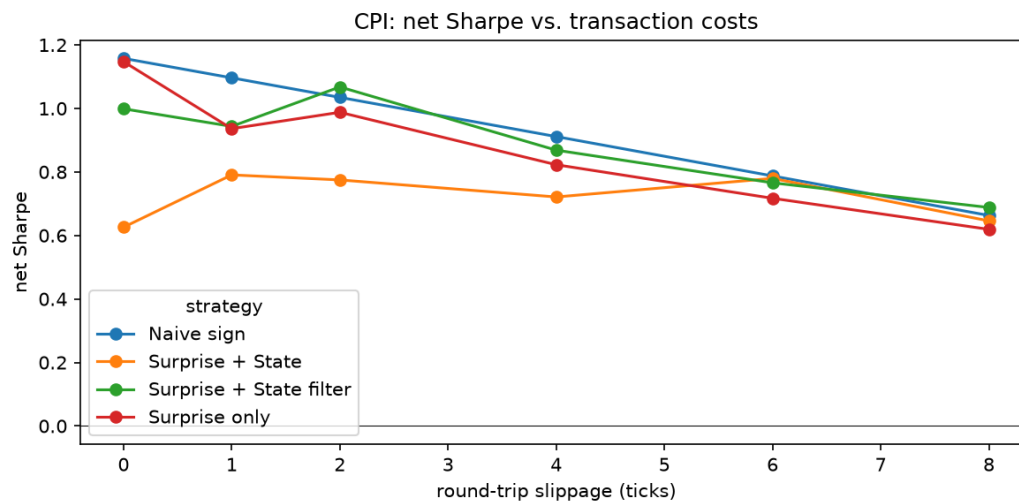


Figure 19. Net Sharpe of the five strategies in the three universes (CPI; CPI + PCE; CPI + PCE + NFP), ES (top), NQ (middle), ZN (bottom).

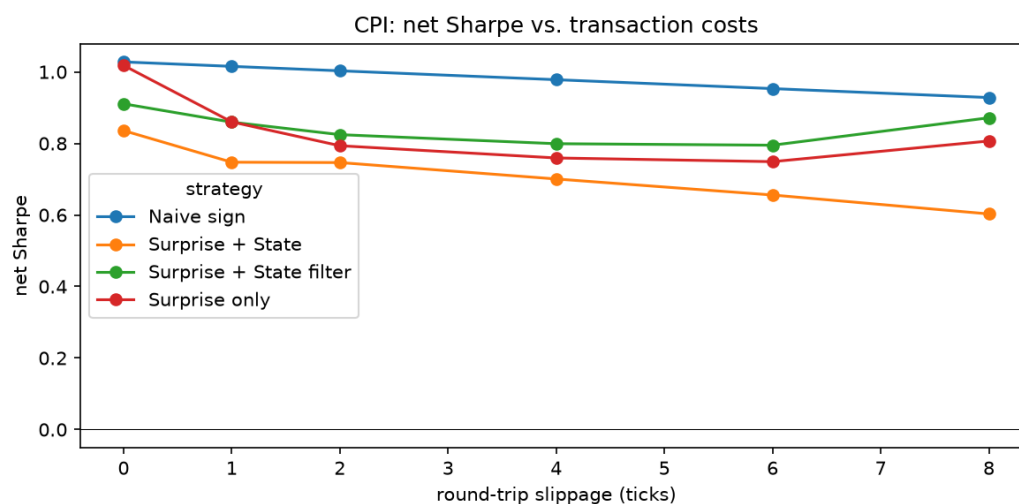
Reading Figure 19 across markets. In ES and NQ the CPI bars are the tallest for every strategy and the bars shrink as PCE and NFP are added: CPI carries the signal. Within CPI, the naive rule and surprise only are at least as high as any state model. In ZN every bar is negative; the naive rule loses the most in the broader universes (-0.65) because it pays the 3 bps cost on every release, and the fitted models' bars rest on only 1-13 trades.

6.4.5 Robustness

ES | E-mini S&P 500



NQ | E-mini Nasdaq-100



ZN | 10-year T-note

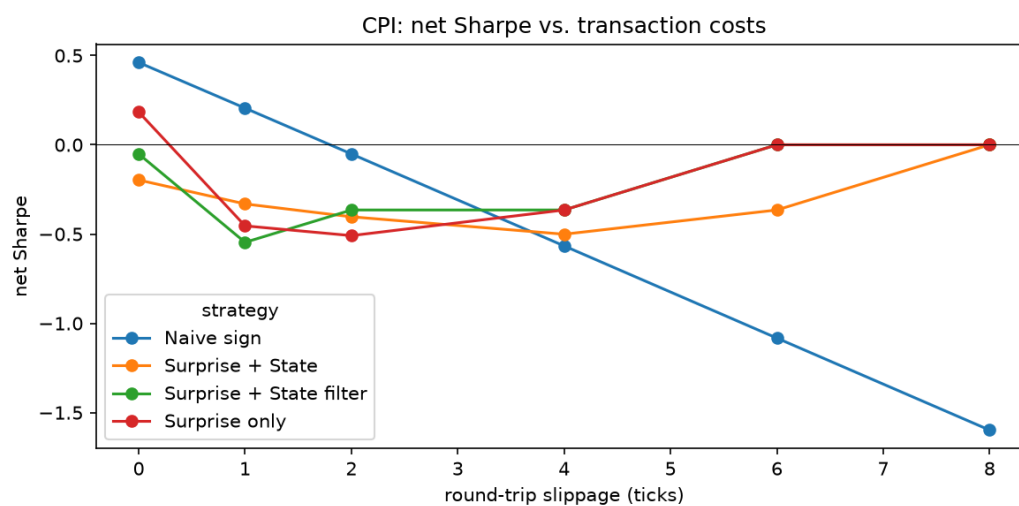


Figure 20. CPI net Sharpe as round-trip slippage rises from 0 to 8 ticks (plus \$4.50 commission), ES (top), NQ (middle), ZN (bottom).

Reading Figure 20 across markets. The lines slope down gently in **ES** (surprise only: 1.15 at 0 ticks, 0.62 at 8 ticks) and are almost flat in **NQ** (0.75–1.02), because one NQ tick is only about 0.1–0.3 bps of the contract value. In **ZN** the naive rule starts positive (0.46 with commission only) and crosses zero at about 2 ticks, and the fitted models trade too rarely for the curve to be meaningful. Cost robustness is therefore a property of the **market's tick size relative to its moves**, not only of the strategy.

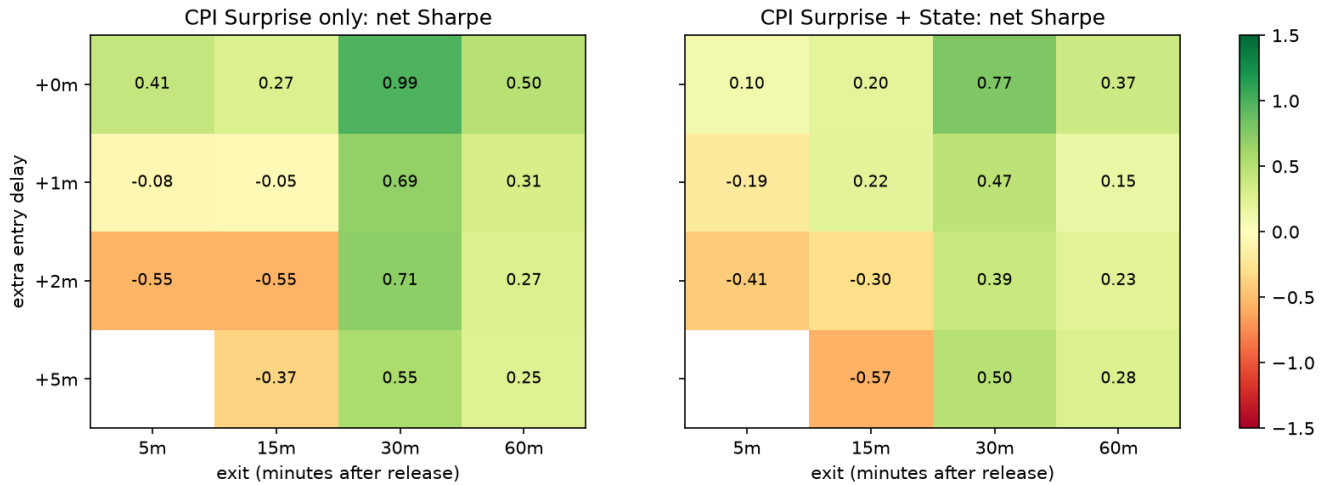


Figure 21. ES: CPI surprise-only Sharpe by entry delay (rows) and holding period (columns).

Robustness (CPI, surprise only)	ES	NQ	ZN
Sharpe at 0 → 8 ticks	1.15 → 0.62	1.02 → 0.81	naive: 0.46 at 0 ticks → -0.05 at 2 ticks all negative
30-min hold, entry delayed 1 / 2 / 5 min	0.55–0.71	0.60 / 0.62 / 0.51	
5-min hold, entry delayed 1 min	collapses	0.43 → 0.05	—
Surprise only ≥ surprise + state in every window/threshold variant	yes	yes	not testable

The edge is a **slow continuation over about 30 minutes**, not a fast reaction: short holds collapse with a one-minute delay, the 30-minute hold does not. In ES and NQ the CPI strategy is robust to four times the base cost.

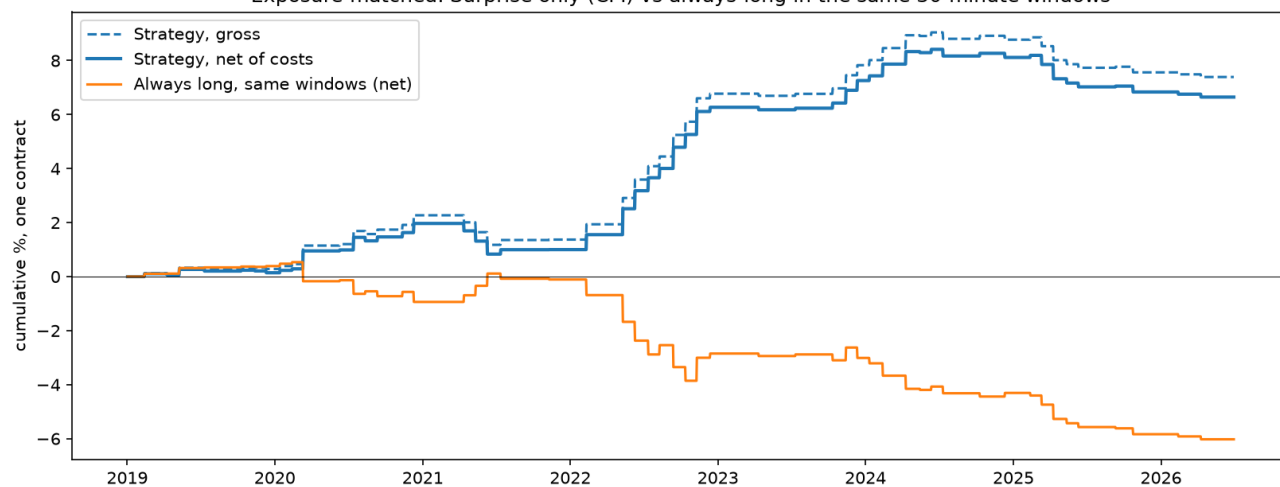
6.4.6 CPI alone vs a broader universe, and the peer format

Net Sharpe, surprise only	CPI	CPI + PCE	CPI + PCE + NFP
ES	0.99	0.71	0.59
NQ	0.79	0.34	0.37

Adding PCE and NFP **dilutes** the CPI signal in both markets.

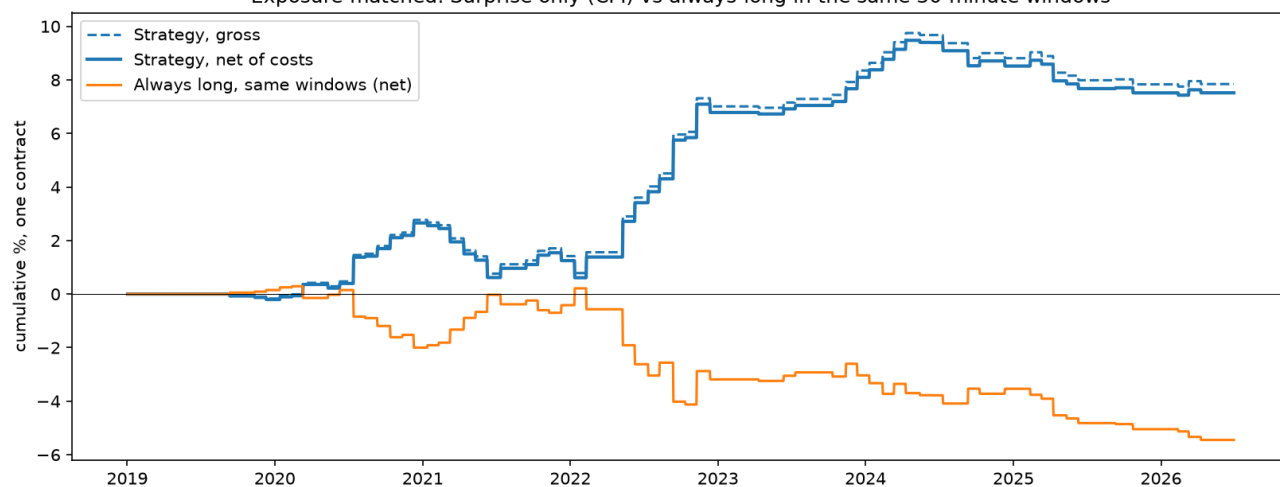
ES | E-mini S&P 500

Exposure-matched: Surprise only (CPI) vs always-long in the same 30-minute windows



NQ | E-mini Nasdaq-100

Exposure-matched: Surprise only (CPI) vs always-long in the same 30-minute windows



ZN | 10-year T-note

Exposure-matched: Surprise only (CPI) vs always-long in the same 30-minute windows

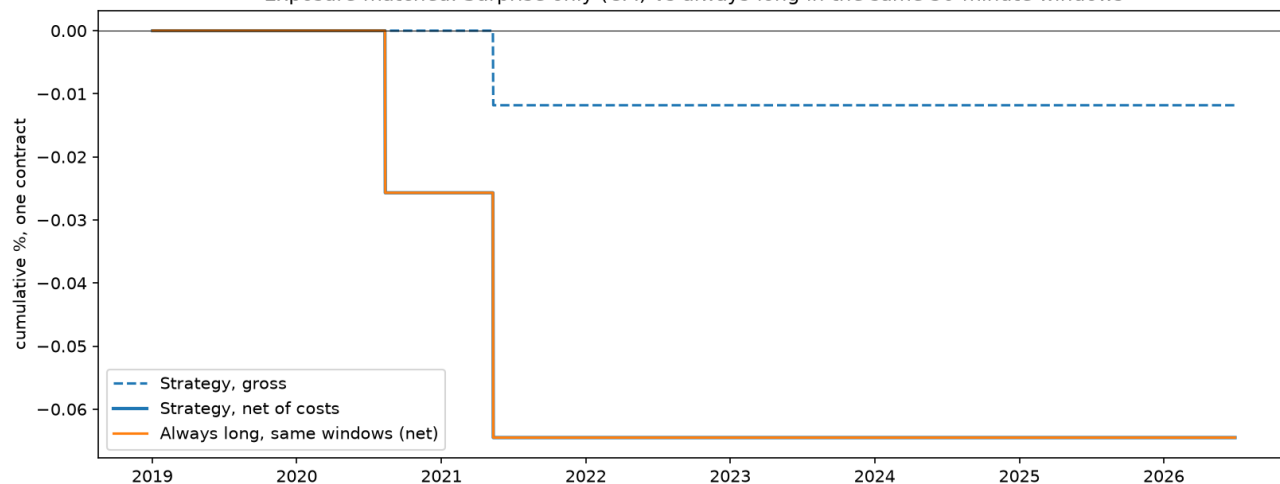


Figure 22. CPI surprise only in the peer (DTW team) chart format: cumulative % of one contract, strategy gross (dashed) and net (solid) vs always-long in the same 30-minute windows (orange), ES (top), NQ (middle), ZN (bottom).

Reading Figure 22 across markets. In ES and NQ the strategy line climbs while the always-long line in the **same** windows falls (ES: +6.6% vs −6.0%; NQ: +7.5% vs −5.4%). Being in the market at CPI time earns nothing by itself; the profit comes entirely from the **direction** of the surprise. The gap between gross and net lines is small, confirming that costs are not the binding constraint for CPI. In ZN the strategy line is flat because the surprise model makes only two trades.

In the peer's daily format, ES CPI surprise only earns a total of 6.6% (daily Sharpe 0.97, max drawdown −1.8%, beta −0.001), while being **always long in the same windows loses 6.0%**: the direction of the surprise is worth about 13 percentage points. NQ is the same (+7.5% vs −5.4%).

6.5 Notebooks 05 and 05b: scaling from CPI to many releases

Question (RQ5). The CPI strategy trades only 7–12 times a year; the peer DTW strategy trades about 185 times a year on ES. If we trade the surprises of many releases, how many trades do we get, and does the edge survive?

Design. All 75 release families; each year the top 10 by the impact screen (history only) are traded, CPI always included. Line signs from the jump regression on history; family signal x = mean of signed z-scores. Sizing: naive sign, per-family surprise model, or Jack's signal $\times$ volatility. Also tested: a second window (T+30 $\rightarrow$ T+60) and a two-window book. Credits: release-family rule, impact screen and volatility sizing from Jack Duncan; $\pm 5\sigma$ winsorisation from Aaryen Mehta.

6.5.1 Which releases does each market react to?

Family rank in the impact screen, 2019–2026	ES	NQ	ZN
CPI	1st–5th (1st from 2023)	1st from 2022	2nd–6th
Employment Report	1st–4th	1st–4th	1st–3rd
FOMC rate decision	high from 2020, but ~no surprise to trade	3rd–4th from 2021	1st every year from 2020
Also regularly admitted	Retail Sales, ISM Services, ISM Manufacturing	ISM Services, ISM Manufacturing, Retail Sales, U. of Michigan	ISM Services, ISM Manufacturing, S&P Global PMIs

The signal is correctly signed out of sample for most families: corr(x , instant jump) in 2019–2026 is **+0.55 (ES), +0.56 (NQ), +0.56 (ZN) for CPI**, and positive for most admitted families.

6.5.2 How many trades, and how profitable?

2019–2026, net of costs	ES trades/yr	ES Sharpe	NQ trades/yr	NQ Sharpe	ZN trades/yr	ZN Sharpe
CPI · naive	12	0.51	12	0.54	12	−0.67
CPI · surprise model	9	0.76	11	0.55	0.1	−0.37
CPI · signal $\times$ vol	12	0.77	12	0.74	12	−0.29
All · naive	98	0.65	97	0.71	82	−2.42
All · surprise model	43	0.11	66	0.37	2	−0.74
All · signal $\times$ vol	101	0.56	100	0.37	84	−1.11
Combined (CPI model + others $\times$ vol)	98	0.40	101	0.23	72	−1.25

2019-2026, net of costs	ES trades/yr	ES Sharpe	NQ trades/yr	NQ Sharpe	ZN trades/yr	ZN Sharpe
All · signal × vol · 2 windows	201	0.16	199	0.00	168	-2.02
Peer: Jack DTW (slide, ES)	185	0.61				

All · naive (the multi-release book)	ES	NQ	ZN
Trading days per year	91	90	—
Net bps per trade	1.7	2.4	-3.4
Total net P&L	12.8%	17.5%	-20.5%
Max drawdown	-4.8%	-5.7%	-20.6%
Sharpe 2019-21 / 2022-26	0.32 / 0.83	0.35 / 0.91	-3.36 / -2.17
Sharpe excluding 2022	0.28	0.44	-2.53
Random-sign p	< 0.001	0.005	0.81
Sharpe at 0 / 1 / 2 / 4 ticks	1.09 / 0.87 / 0.65 / 0.20	0.82 / 0.77 / 0.71 / 0.60	-0.56 / -1.49 / -2.42 / -4.24
Second window alone (T+30 → T+60)	-0.79	-0.23	-3.08

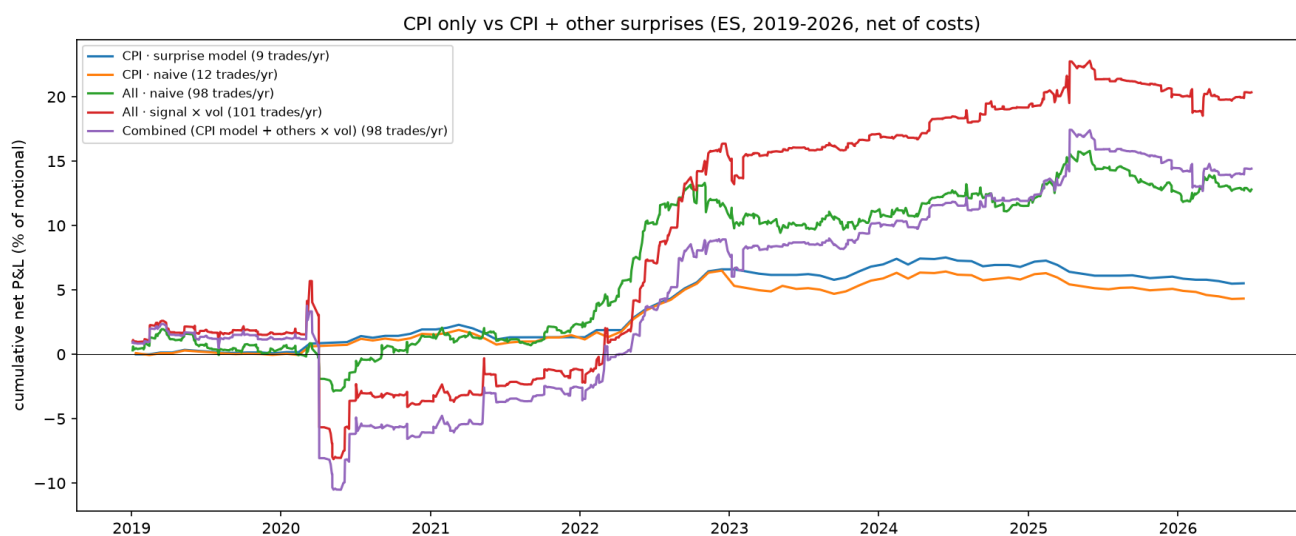


Figure 23. ES: cumulative net P&L of CPI-only and multi-release books.

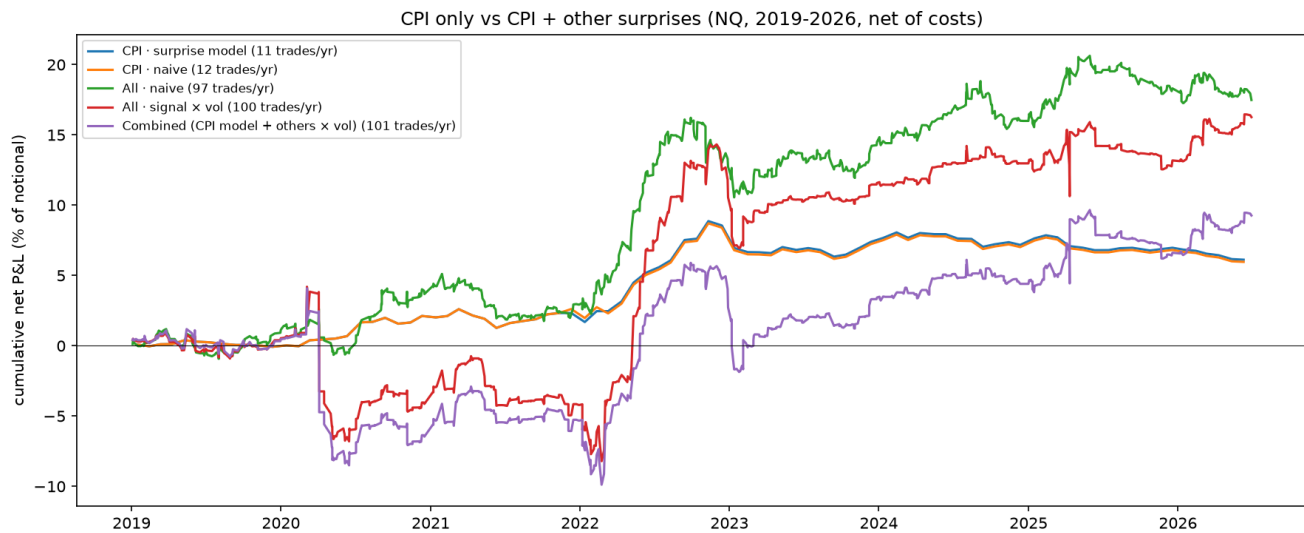


Figure 24. NQ: the same books.

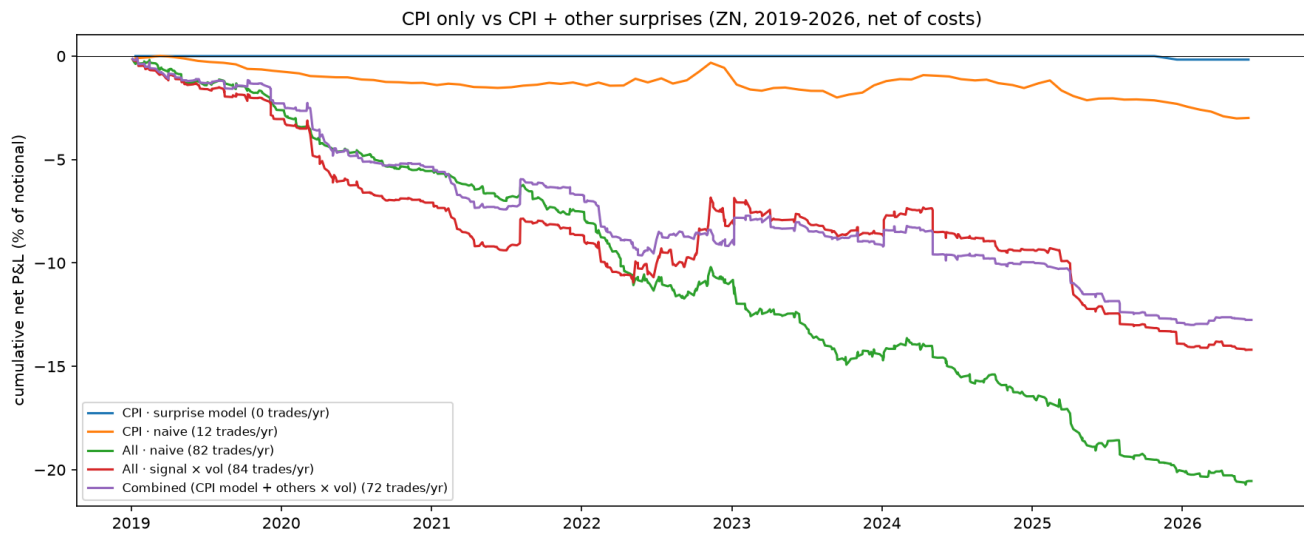


Figure 25. ZN: every book loses money after costs.

6.5.3 Which releases earn money?

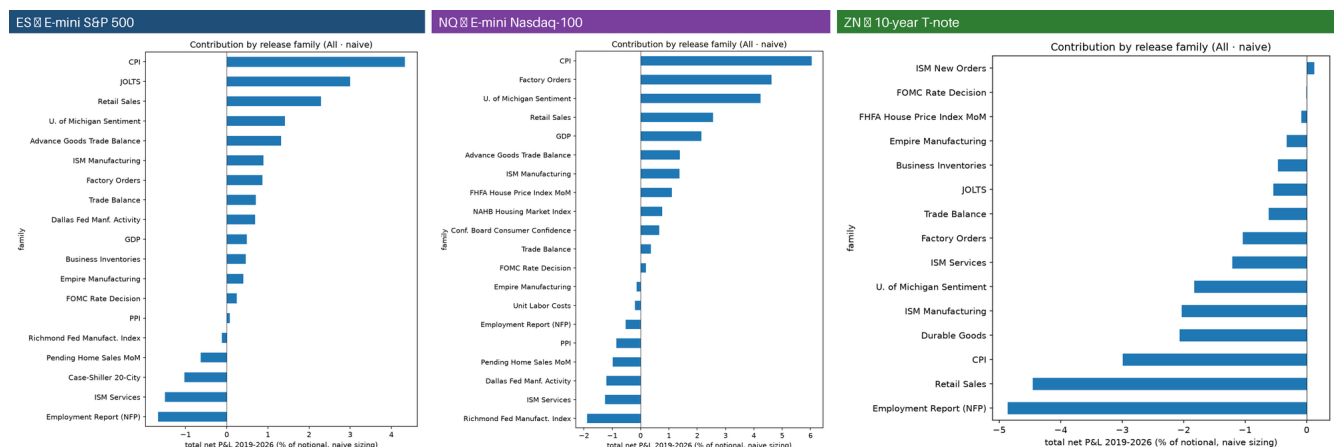


Figure 26. Total net P&L of the multi-release naive book by release family, 2019–2026 (% of notional, one contract per trade), ES (left), NQ (centre), ZN (right).

Reading Figure 26 across markets. In the equity markets the bars are mostly to the right, and **CPI is the largest contributor in both** (ES +4.3%, NQ +6.0%). Behind CPI the order differs (ES: JOLTS, Retail Sales, U. of Michigan; NQ: Factory Orders, U. of Michigan, Retail Sales, GDP), which is itself a warning: apart from CPI, no family's contribution is stable across the two indices, so the broad book should be seen as diversification across many small, noisy edges. Both equity books lose on the **Employment Report** and **ISM Services**, the two releases whose several components often point in opposite directions. In ZN almost every bar is to the left, and the largest losses are on the releases ZN reacts to most (Employment Report −4.9%, Retail Sales −4.5%, CPI −3.0%): the bond market reprices them in the first minute, and each trade afterwards pays about 3 bps for nothing.

- **ES:** CPI +4.3%, JOLTS +3.0%, Retail Sales +2.3%, U. of Michigan +1.4%; Employment Report −1.7%, ISM Services −1.5%.
- **NQ:** CPI +6.0%, Factory Orders +4.6%, U. of Michigan +4.2%, Retail Sales +2.6%, GDP +2.1%; Richmond Fed −1.9%, ISM Services −1.3%.
- **ZN:** every family with more than 4 trades loses about the cost on every trade.

No family except CPI comes close to individual significance: the multi-release book works through **diversification across many small edges**.

6.5.4 An important finding: the definition of the CPI surprise

The ES CPI naive book in Notebook 05 (Sharpe 0.51) is weaker than in Notebook 04 (1.03) **with identical trade returns**. Notebook 04 uses the headline CPI MoM surprise; Notebook 05 averages all eight CPI lines. The two disagree in direction on **20 of 86 releases**, mostly months where the headline surprise was almost exactly zero ($|z| < 0.2$), and those 20 trades swing P&L by about 4.3 percentage points (one release, January 2023, is worth +115 bps under one definition and −118 bps under the other). The same gap appears in NQ (1.00 vs 0.54). The surprise-model and volatility-sized versions, which trade little on small surprises, are more stable.

6.5.5 Interpretation

- **The trade count can be raised tenfold.** About 10 screened releases give ~100 trades a year per market, against ~10 for CPI alone.
- **The edge survives in equity futures.** Sharpe 0.65 (ES) and 0.71 (NQ), comparable to the peer DTW strategy (0.61) with a larger edge per trade (1.7–2.4 vs 1.2 bps) and an economically explained signal.
- **More releases bring less dependence on 2022**, especially in NQ (ex-2022 Sharpe 0.44).
- **Costs matter more for the broad book.** In ES its Sharpe falls to 0.20 at 4 ticks; NQ is far more robust (0.60) because a tick is a smaller share of the NQ price.
- **The effect lasts about 30 minutes.** A second window loses money in every market, so the honest ceiling is about **100 trades a year per market** at one trade per release.
- **The simplest sizing works best.** Per-family surprise models have too little history (~24 releases per family) and volatility sizing raises drawdowns.

6.6 Notebooks 06 and 06b: surprise plus DTW pattern, and the five-window ladder

Question (RQ6). Does the shape of the price path around a release add information beyond the surprise? And can trading five consecutive windows after each release (the peer's structure) reach ~185 trades a year profitably?

Design. The peer DTW specification (Section 3.2) on the same releases, entry, exit, costs and window as Notebook 05. Four combinations: surprise only, DTW only, surprise + DTW agree, surprise

+ DTW blend. Then the **ladder**: five windows after each release ($T+1 \rightarrow T+30$, ..., $T+120 \rightarrow T+150$), with DTW rebuilt for each window.

6.6.1 Does DTW predict the 30-minute return?

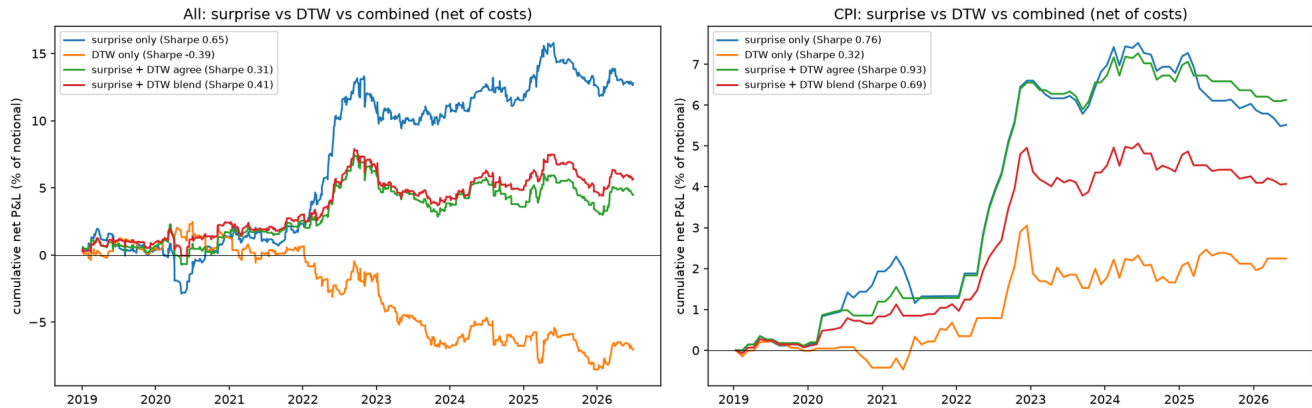
DTW, all releases 2019-2026	ES	NQ	ZN
Releases with a DTW signal	5,181	5,229	5,244
Rank IC with the 30-min return	+0.010	+0.026	+0.039
Hit rate of the sign	49.2%	50.9%	46.6%
corr(DTW, surprise)	0.11	0.07	0.02

A useful directional signal in this setting would show a rank IC of about 0.05-0.10, stable across years. DTW is close to zero in every market and unrelated to the surprise.

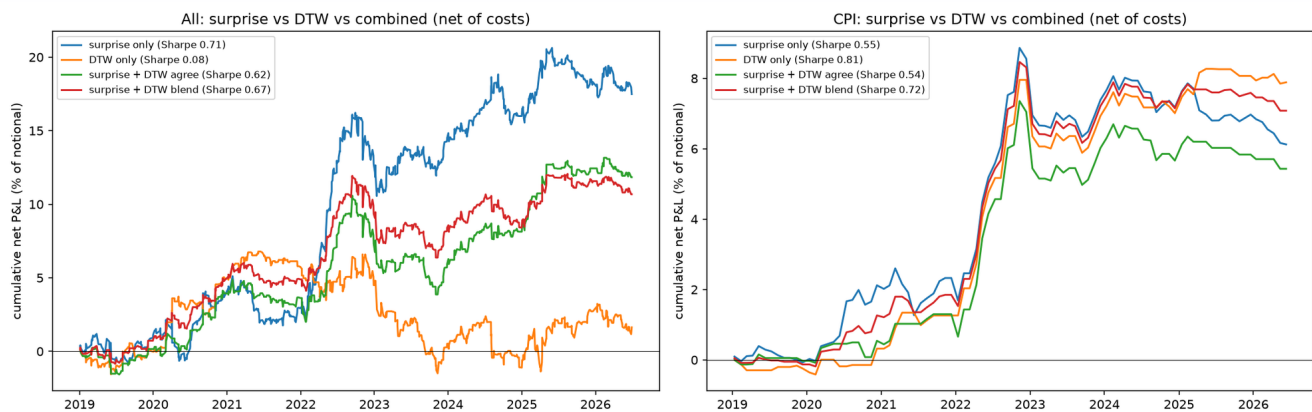
6.6.2 Does adding DTW improve the surprise strategy?

Net Sharpe, ~10 releases	ES	NQ	ZN
Surprise only	0.65	0.71	-2.42
Surprise + DTW agree	0.31	0.62	-1.57
Surprise + DTW blend	0.41	0.67	-1.51
DTW only	-0.39	0.08	-1.25
CPI: surprise model $\rightarrow$ + DTW agree	0.76 $\rightarrow$ 0.93 (P(not better) 0.21)	0.55 $\rightarrow$ 0.54	—
CPI: DTW only	0.32	0.81 (not significant)	-0.55

ES | E-mini S&P 500



NQ | E-mini Nasdaq-100



ZN | 10-year T-note

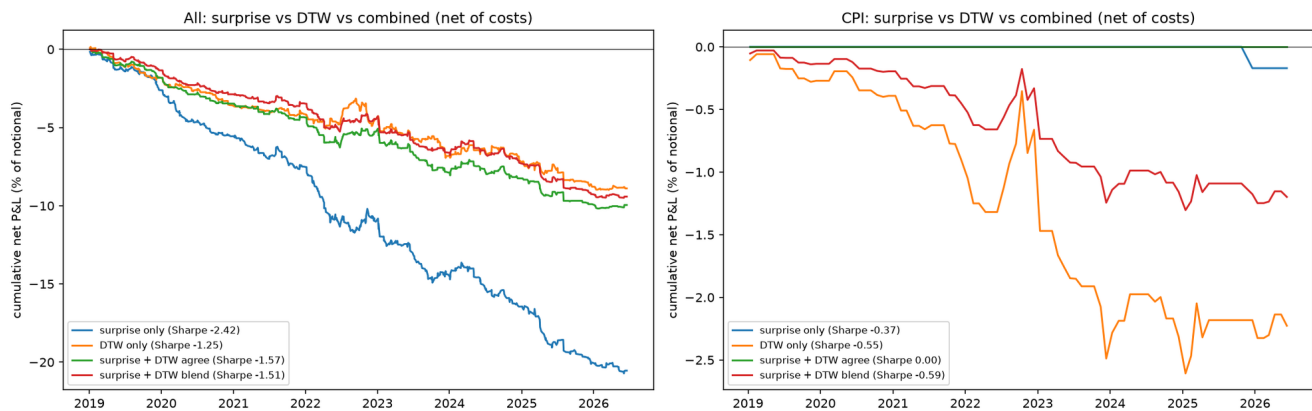


Figure 27. Cumulative net P&L of surprise only, DTW only, surprise + DTW agree and surprise + DTW blend, for the ~10-release universe (left) and CPI (right), ES (top), NQ (middle), ZN (bottom). Sharpe ratios in the legends.

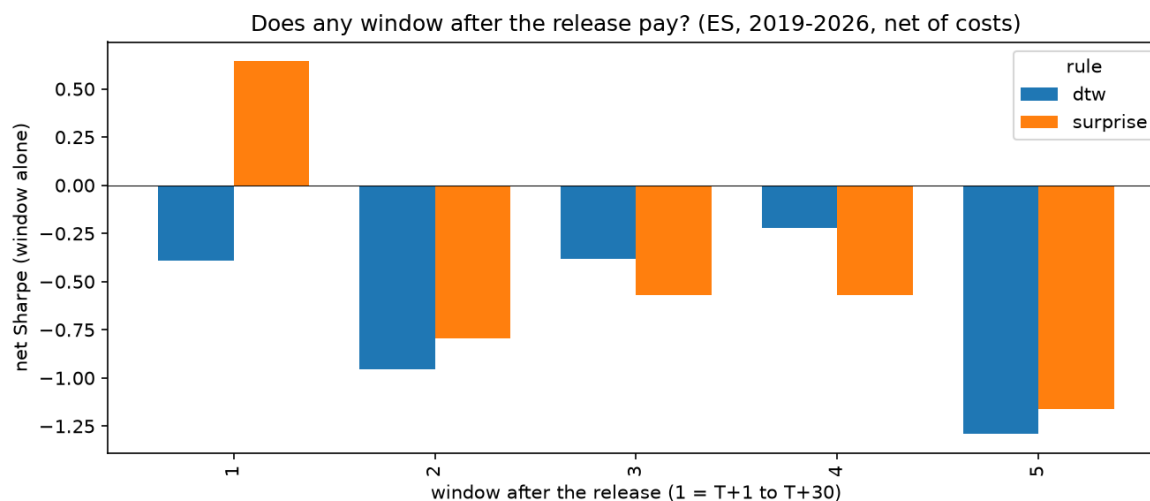
Reading Figure 27 across markets. In **ES** (all releases) the surprise-only line rises in steps, mostly in 2022, while the DTW-only line is flat until 2022 and then falls steadily to about -7%; the two combinations sit between them, closer to zero. In **NQ** DTW only actually rises in 2019-2022 (it was positive in the first half of the sample) but gives it all back from 2023, a typical sign of a pattern that does not persist; again surprise only is the highest line. For CPI the four lines move together, because CPI's surprise and DTW agree on most of the large 2022 moves; the differences between them are small and point in different directions in ES and NQ. In **ZN** every line falls almost linearly: the loss is essentially the cost paid on each trade, and the surprise line falls fastest because it trades

most.

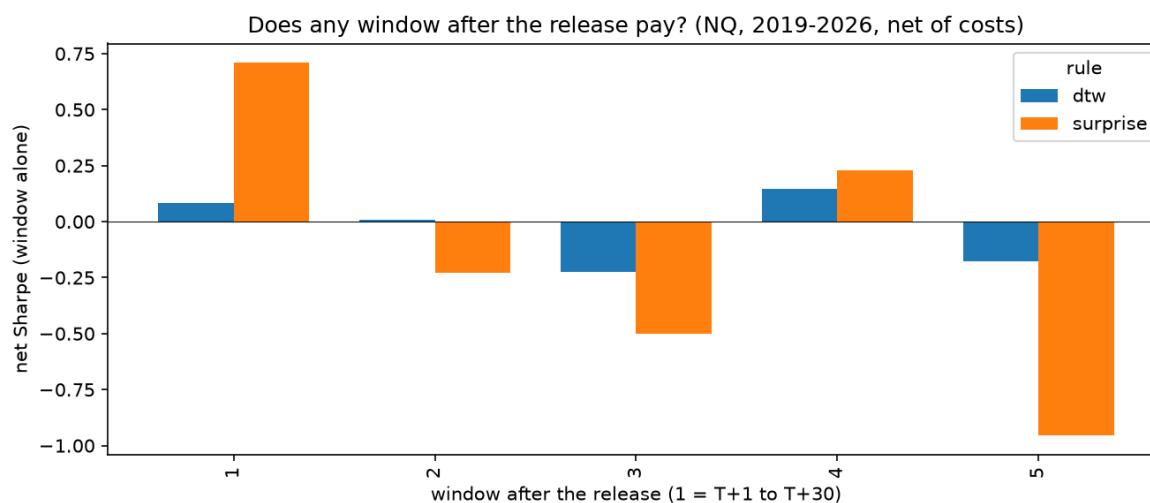
- In ES and NQ **every combination lowers the Sharpe** of the multi-release surprise book.
- For CPI the results **point in different directions in different markets** (ES: the agreement filter looks best; NQ: DTW alone looks best), and none is significant. We treat them as noise.
- In ZN, DTW books lose less than the surprise only because they trade less and pay less cost.
- DTW does not separate good surprise trades from bad ones in ES (trades where DTW disagreed earned *more*, +2.4 vs +1.1 bps).

6.6.3 The ladder: can we reach ~185 trades a year?

ES | E-mini S&P 500



NQ | E-mini Nasdaq-100



ZN | 10-year T-note

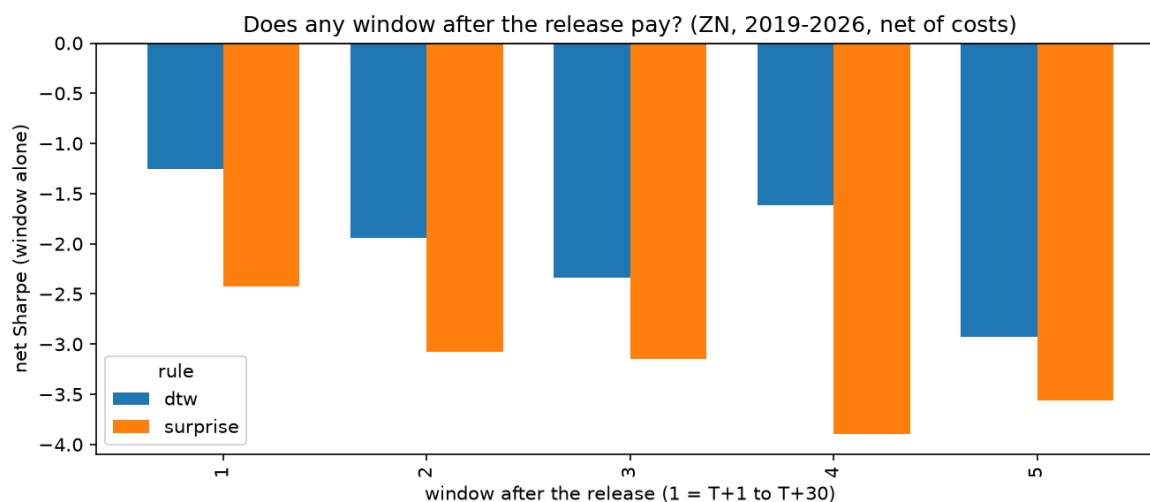


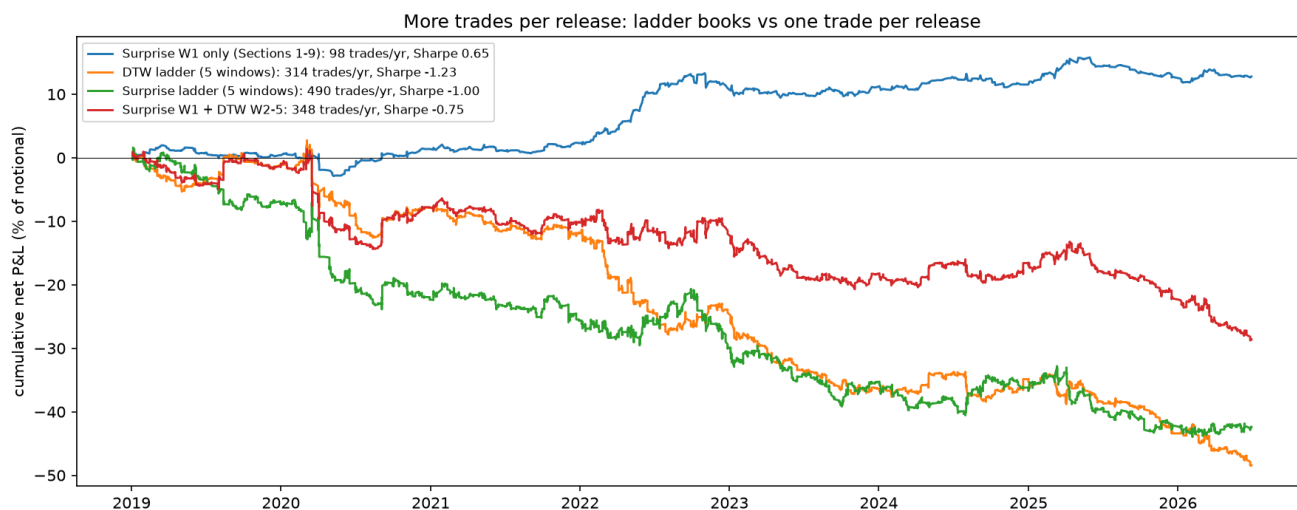
Figure 28. Net Sharpe of the surprise signal (orange) and the DTW signal (blue) traded alone in each of the five 30-minute windows after the release, ES (top), NQ (middle), ZN (bottom).

Reading Figure 28 across markets. The ES and NQ panels have the same shape: **only window 1 is clearly positive, and only for the surprise** (ES +0.65, NQ +0.71). From window 2 onwards the surprise bars are negative in ES and mostly negative in NQ, and the DTW bars are near zero or negative in every window. NQ's small positive window 4 (+0.23 surprise, +0.15 DTW) has a rank IC of about 0.02 and is best read as noise. In ZN every bar is deeply negative (−1.2 to −3.9): no window pays for either signal.

Surprise rank IC by window	ES	NQ	ZN
W1: T+1 → T+30	+0.139	+0.088	−0.026
W2: T+30 → T+60	+0.008	−0.014	−0.011
W3: T+60 → T+90	−0.055	−0.056	+0.010
W4: T+90 → T+120	+0.001	+0.022	−0.044
W5: T+120 → T+150	+0.005	−0.041	+0.038

Ladder books: trades/yr, Sharpe	ES	NQ	ZN
Surprise, window 1 only	98, +0.65	97, +0.71	82, −2.42
Surprise W1 + DTW W2-5	348, −0.75	342, +0.27	294, −5.35
Surprise ladder (5 windows)	490, −1.00	484, −0.29	410, −7.65
DTW ladder (5 windows)	314, −1.23	309, −0.07	267, −3.92
Peer: Jack DTW (slide, ES)	185, +0.61		

ES | E-mini S&P 500



NQ | E-mini Nasdaq-100



ZN | 10-year T-note

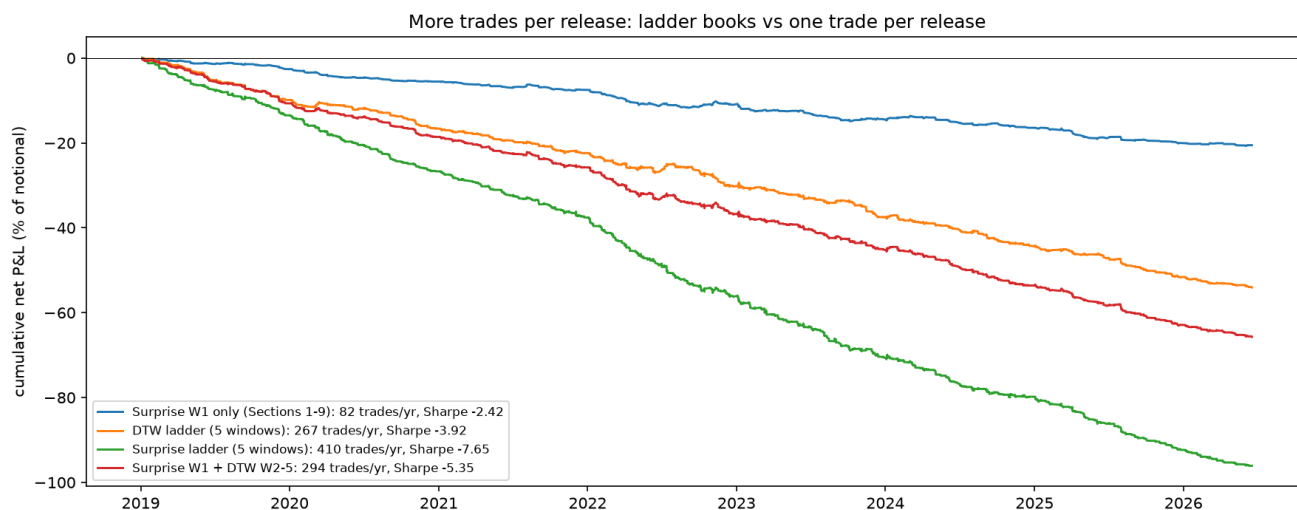


Figure 29. Cumulative net P&L of the three ladder books (DTW ladder, surprise ladder, surprise W1 + DTW W2-5) against window 1 only, ES (top), NQ (middle), ZN (bottom).

Reading Figure 29 across markets. In **ES** window 1 alone is the only line that ends positive (+12.8%); every ladder book falls steadily, to between −29% and −48%, because each extra window adds about 1–2 bps of cost with no edge. In **NQ**, where the cost is below 1 bps, the ladder books lose much less and “surprise W1 + DTW W2-5” even ends positive (+12.9%), but it is still below window 1 alone (+17.5%) with a drawdown three times larger. In **ZN** every ladder book loses between 54% and 96% of notional. The chart makes the trade-count trade-off visible: more windows per release add turnover and cost, not profit.

Interpretation. The surprise’s information is strong in the first 30 minutes (rank IC 0.14 in ES) and **gone afterwards** in every market. DTW has no information in any window. The ladder books trade 270–490 times a year, more than the peer’s 185, but every added window lowers the Sharpe: the trade count can be raised only by adding losing (or, in NQ, weaker) trades.

6.7 Notebook 07: the multi-market portfolio

Question (RQ7a). Does combining ES, NQ and ZN improve the surprise strategy?

Design. Notebooks 05 and 06 run end-to-end in each market (own impact screen, own line signs, own DTW), then combined with equal-risk weights estimated from history only (inverse standard deviation of each market’s 30-minute trade return, ES = 1; ZN receives a weight of about 2.7).

2019–2026, net of costs	Trades / yr	Net bps / trade	Total net P&L	Sharpe	Sharpe ex-2022	Max DD	Random- sign p
ES · All · naive	98	1.7	12.8%	0.65	0.28	−4.8%	< 0.001
NQ · All · naive	97	2.4	17.5%	0.71	0.44	−5.7%	0.005
ZN · All · naive	82	−3.3	−20.5%	−2.42	−2.53	−20.6%	0.81
ES + NQ · All · naive	195	1.7	25.2%	0.72	0.38	−7.4%	< 0.001
ES + NQ · CPI · surprise model	20	6.4	9.9%	0.69	0.02	−3.3%	0.023
3-market · All · naive	277	−1.4	−29.7%	−0.75	−1.17	−31.1%	
3-market · CPI · surprise model	21	6.1	9.4%	0.65	−0.03	−3.8%	0.019
3-market · CPI · surprise + DTW agree	14	9.7	10.0%	0.78	0.08	−2.4%	0.009
Peer: Jack DTW, ES (slide, 2018–26)	185	1.2	18.6%	0.61		−4.9%	

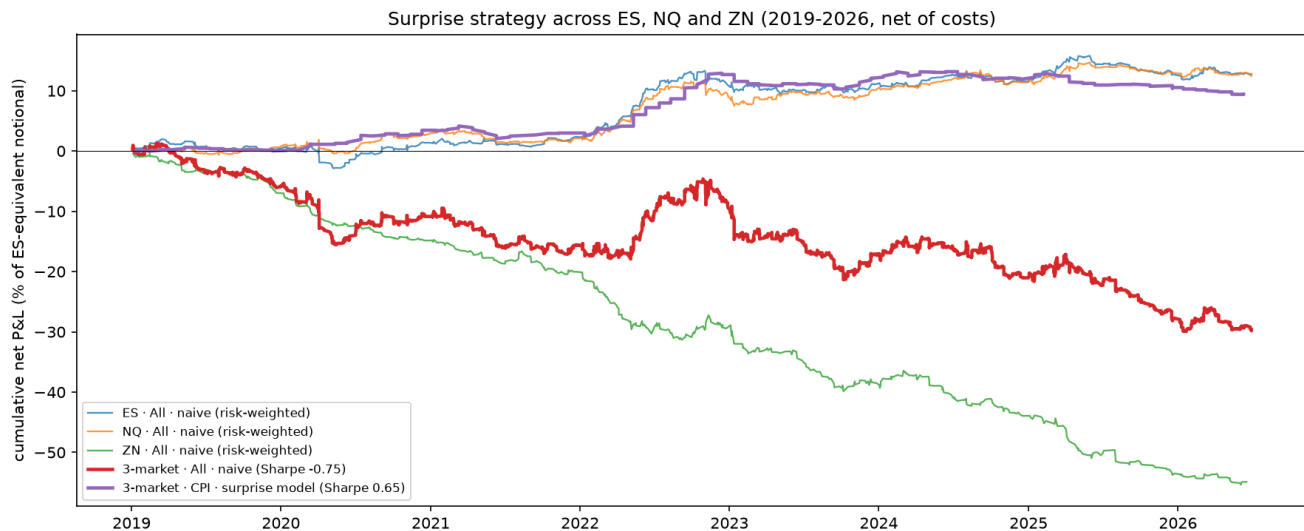


Figure 30. Cumulative net P&L of the single-market and multi-market books.

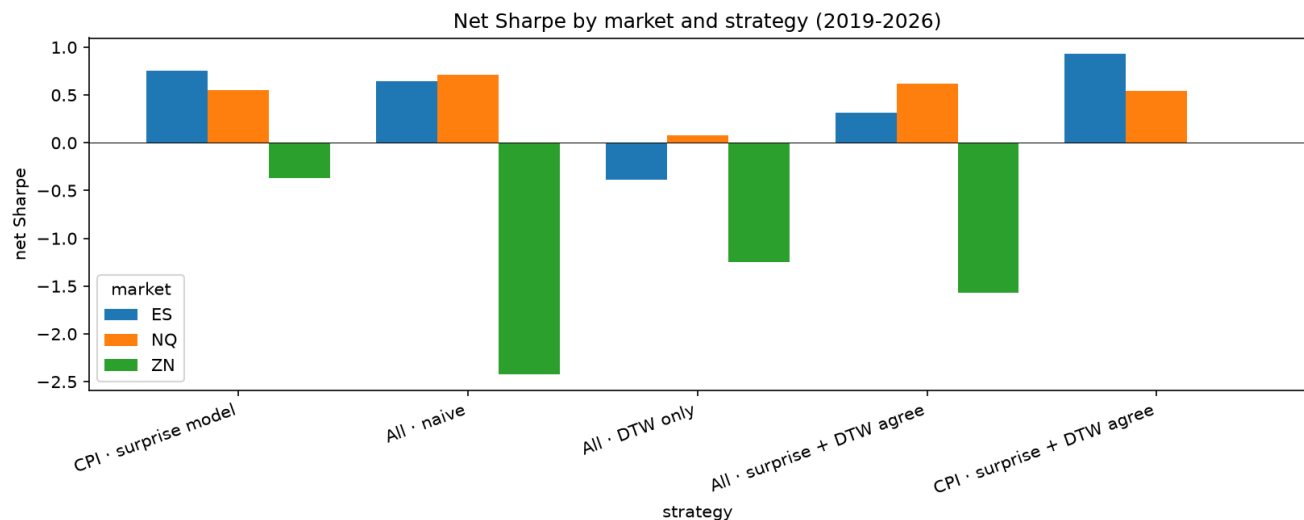


Figure 31. Net Sharpe of each strategy by market.

At the peer's cost assumption ($\frac{1}{4}$ tick per side, no commission):

Sharpe	ours: 2 ticks + \$4.50	$\frac{1}{4}$ tick per side	gross
ES · All · naive	0.65	1.06	1.17
NQ · All · naive	0.71	0.84	0.87
ES + NQ · All · naive	0.72	1.01 (total 35.8%)	1.09
ZN · All · naive	-2.42	-0.76	-0.30

Interpretation. ES and NQ combine well: **~195 trades a year, Sharpe 0.72, total +25%**, above the peer's trade count with a higher Sharpe. Adding ZN destroys the portfolio: it has no edge, its risk weight is the largest, and it loses even gross of costs. The directional signs are economically consistent across markets (a hot CPI print pushes all three down; strong payrolls push ES and NQ up and ZN down), but in ZN the direction is fully priced in the first minute.

6.8 Notebook 08: the four team workstreams at equal costs

Question (RQ7b). The four team members reported results with different cost assumptions, windows, entry times and Sharpe methods. How do they compare on equal terms?

Design. Each workstream's **own trades** are re-priced under four cost scenarios (gross; the peer's $\frac{1}{4}$ tick per side; $\frac{1}{2}$ tick per side; our 2 ticks + \$4.50), on the same window (2019–2026) and with the same Sharpe method. Jack Duncan's trade log is not in the repository, so his ES DTW ladder is **rebuilt from its specification** (`src/team_replication.py`); the rebuild gives 310 trades/yr and Sharpe 0.16 on 2018–2026 at his cost, against his slide's 185 and 0.61, so it is an approximation of his strategy, not his exact result. Aaryen Mehta's classifier is shown both **as reported** (entry at the pre-release price, lag 0) and **executable** (entry one minute later, lag 1).

Net Sharpe (monthly), 2019–2026	Trades / yr	Gross	$\frac{1}{4}$ tick per side	$\frac{1}{2}$ tick per side	2 ticks + \$4.50
Reza · ES · all releases (surprise)	98	1.17	1.06	0.95	0.65
Reza · ES + NQ · all releases	195	1.09	1.01	0.94	0.72
Reza · NQ · all releases	97	0.87	0.84	0.82	0.71
Reza · ES · CPI surprise model	9	0.88	0.85	0.83	0.76
Aaryen · 30 min, as reported (lag 0)	55	2.04	2.03	2.01	1.97
Aaryen · 30 min, executable (lag 1)	55	1.08	1.03	0.98	0.84
Aaryen · 10 min, executable	55	0.32	0.25	0.17	−0.04
Hoshea · ES · DTW event games	372	0.44	0.24	0.04	−0.50
Jack · ES · DTW 5-window ladder (rebuild)	315	0.31	0.10	−0.11	−0.68
<i>Jack · slide (2018–26, his data and costs)</i>	<i>185</i>		<i>0.61</i>		

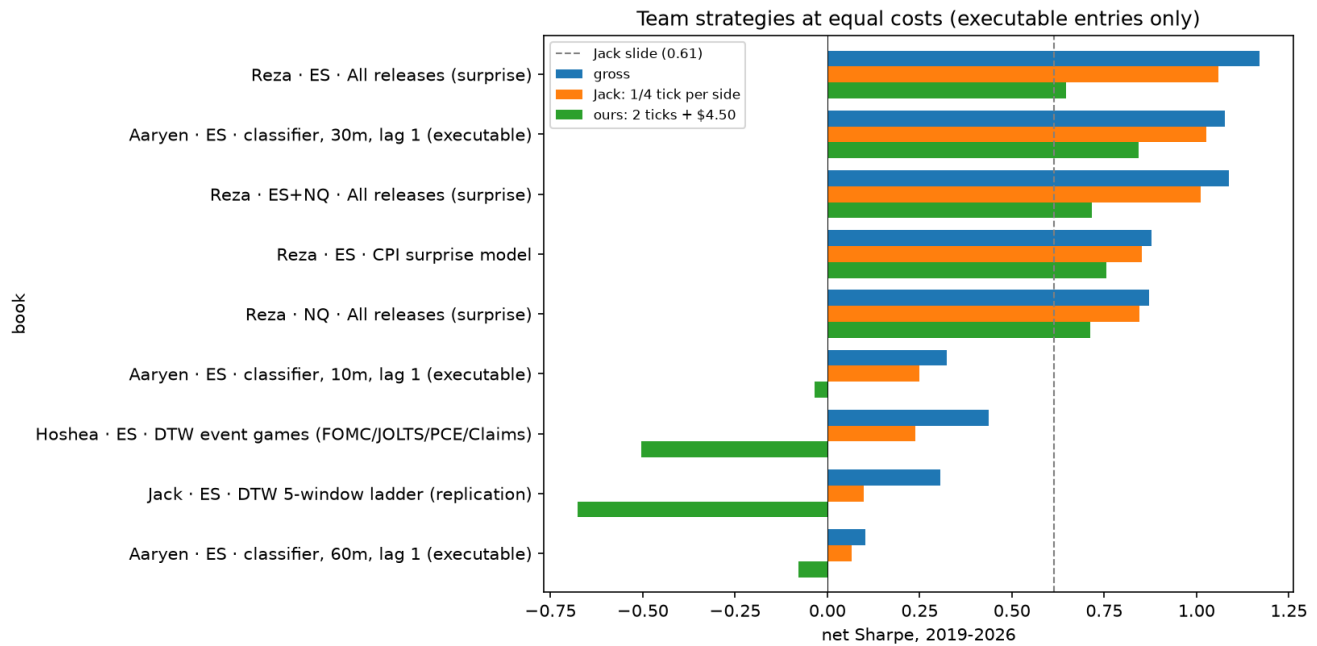


Figure 32. Net Sharpe of every executable workstream under three cost levels.

6.8.1 All four together, three cost levels

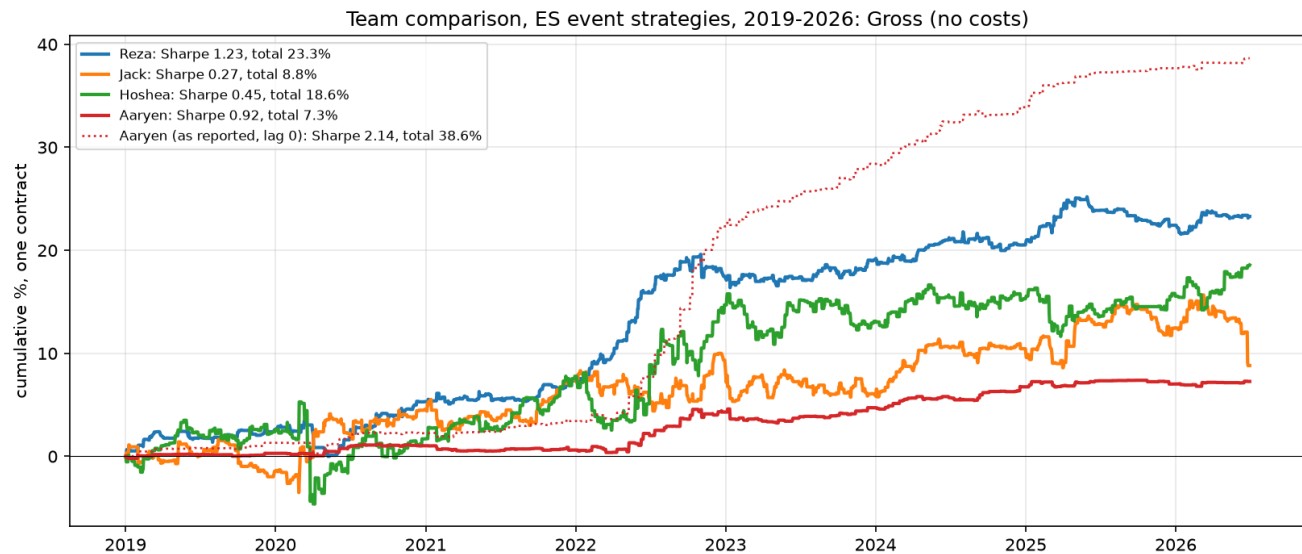


Figure 33. Cumulative P&L (one contract), gross of costs: Reza (ES, all releases), Jack (rebuild), Hoshea, Aaryen (executable; dotted: as reported).

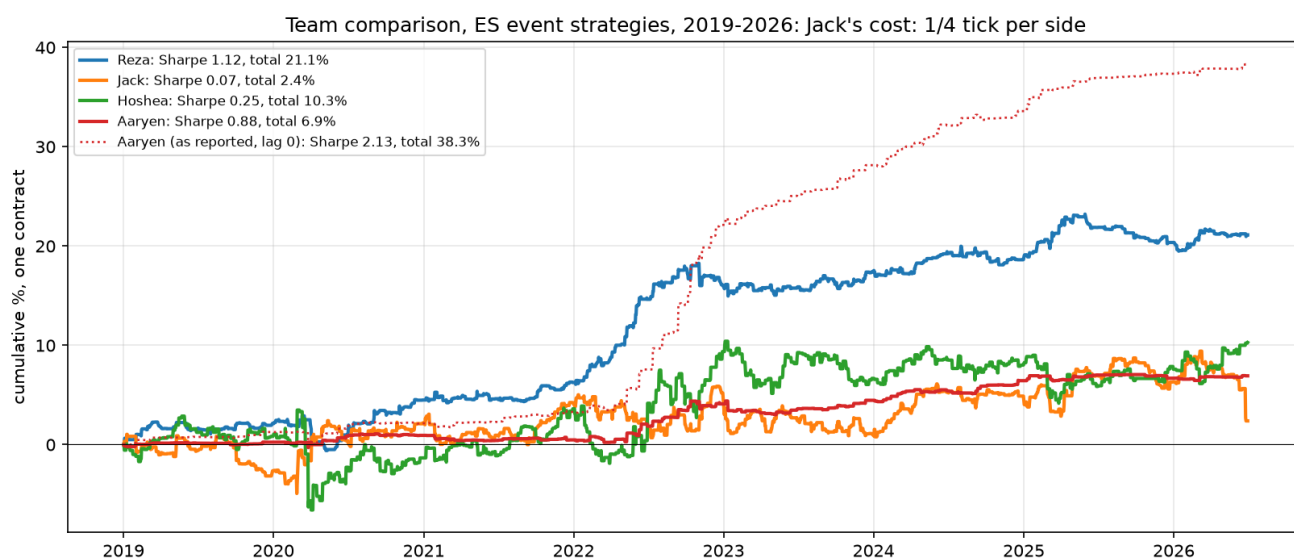


Figure 34. The same at 1/4 tick per side.

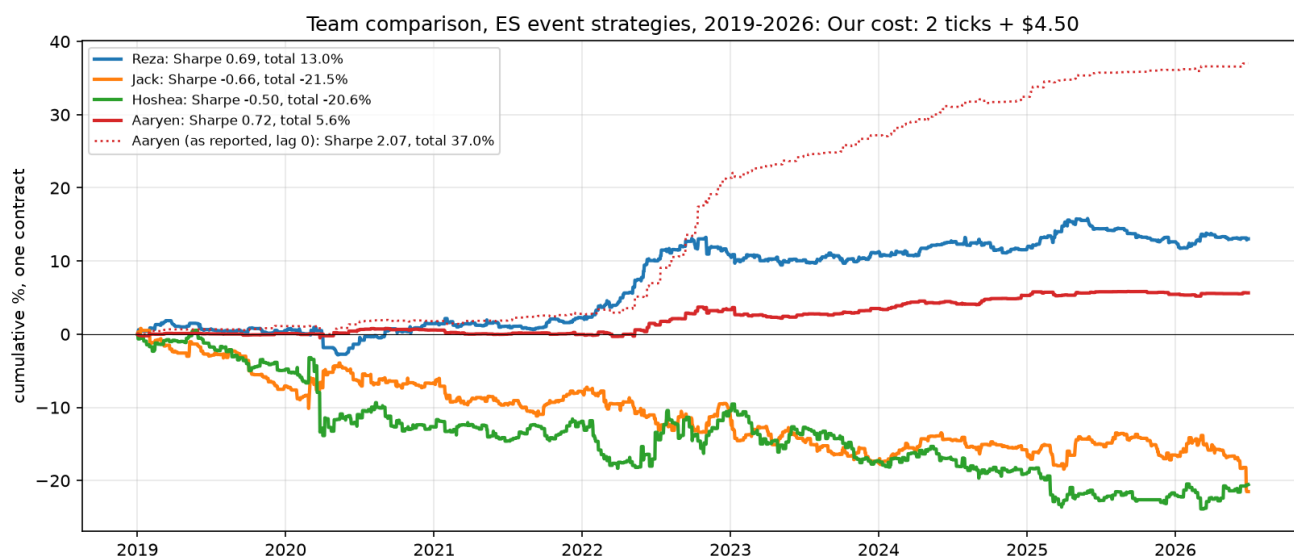


Figure 35. The same at 2 ticks + \$4.50.

Daily Sharpe ($\sqrt{252}$) / total %	Gross	1/4 tick per side	2 ticks + \$4.50
Reza (ES, all releases)	1.23 / 23.3%	1.12 / 21.1%	0.69 / 13.0%
Jack (rebuild)	0.27 / 8.8%	0.07 / 2.4%	-0.66 / -21.5%
Hoshea	0.45 / 18.6%	0.25 / 10.3%	-0.50 / -20.6%
Aaryen (executable)	0.92 / 7.3%	0.88 / 6.9%	0.72 / 5.6%
Aaryen (as reported)	2.14 / 38.6%	2.13 / 38.3%	2.07 / 37.0%

6.8.2 One workflow at a time, in the peer's slide format

The notebook also shows each workflow in the format of the peer's slide: net, net at our cost, gross, always-long in the same windows, and buy-and-hold.

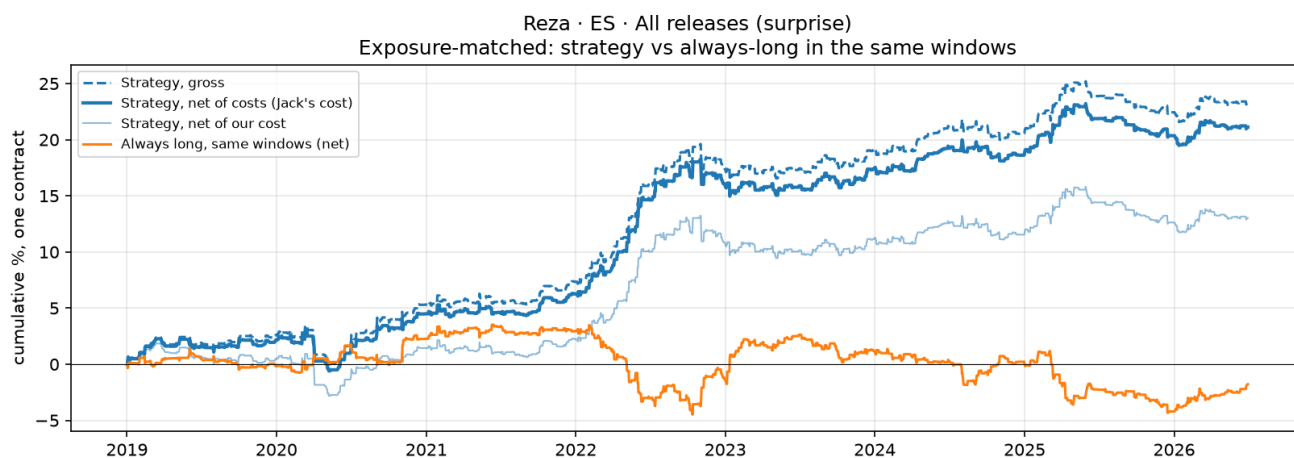


Figure 36. Reza, ES all releases: strategy vs always-long in the same windows.

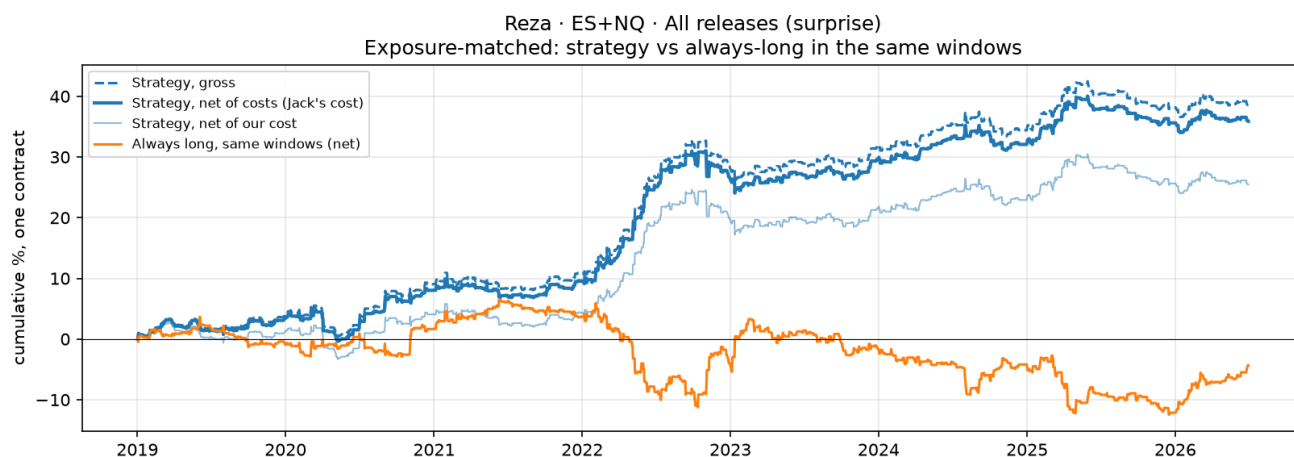


Figure 37. Reza, ES + NQ all releases.

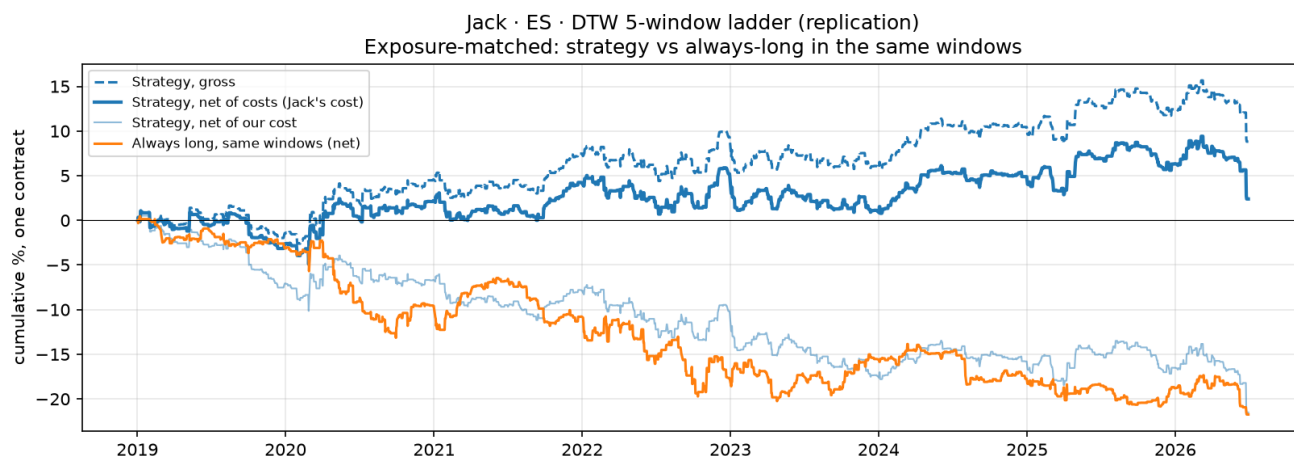


Figure 38. Jack, DTW ladder (rebuild).

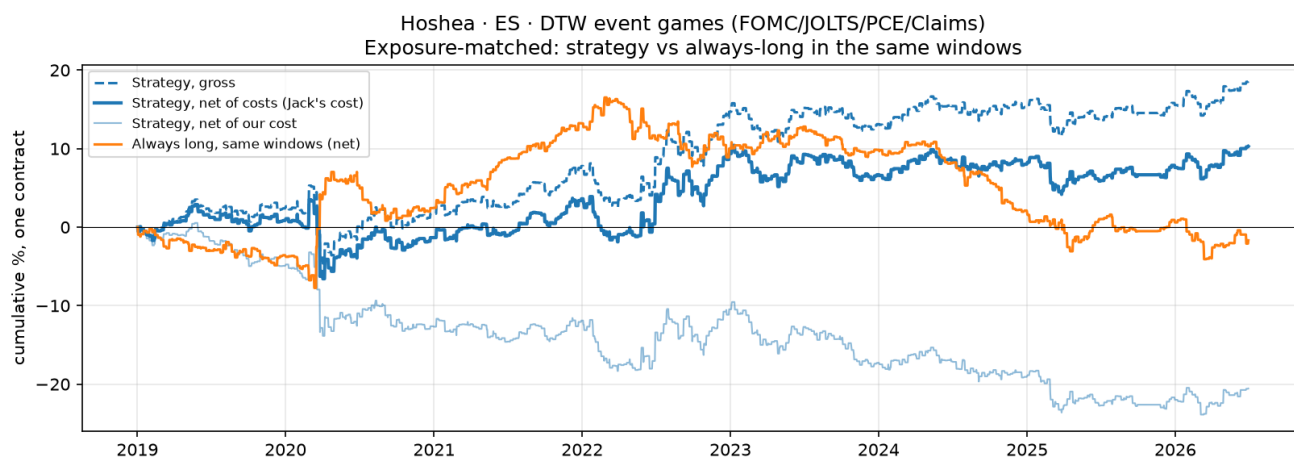


Figure 39. Hoshea, DTW event games.

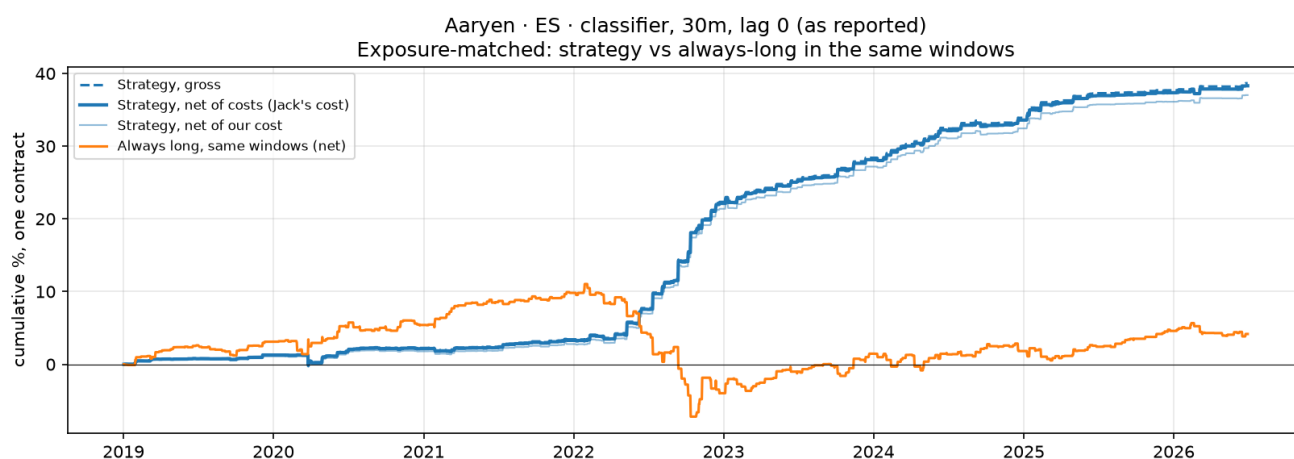


Figure 40. Aaryen, 30-minute classifier, as reported (entry at the pre-release price).

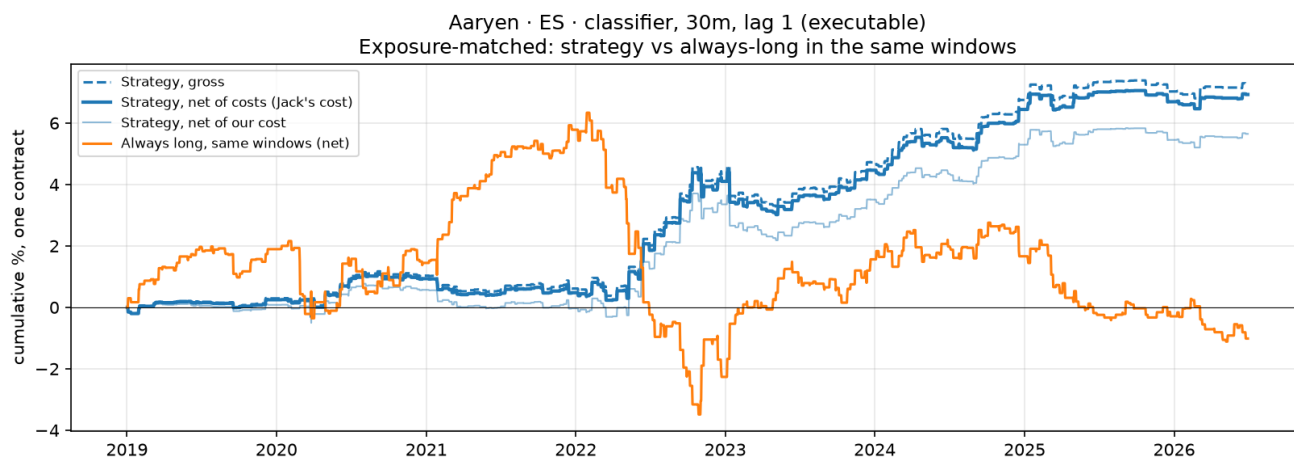


Figure 41. Aaryen, 30-minute classifier, executable (entry one minute later).

Slide format, ¼ tick per side, 2019–2026	Total %	Vol %	Daily Sharpe	Max DD	Beta	Trades / yr	Always long, same windows
Reza · ES · all releases	21.1	2.6	1.12	−3.7%	−0.001	98	−1.8%
Reza · ES + NQ	35.9	4.7	1.06	−7.0%	0.002	195	−4.3%
Jack · rebuild	2.4	4.5	0.07	−7.0%	0.005	315	−21.7%
Hoshea	10.3	5.7	0.25	−10.1%	0.010	372	−1.7%
Aaryen · executable	6.9	1.1	0.88	−1.4%	−0.004	55	−1.0%
Aaryen · as reported	38.3	2.5	2.13	−1.5%	−0.006	55	+4.2%
<i>Jack · slide (his data, 2018–26)</i>	<i>18.6</i>	<i>3.0</i>	<i>0.61</i>	<i>−4.9%</i>	<i>−0.005</i>	<i>185</i>	

Interpretation.

- **Entry timing explains the largest difference in the team.** Aaryen’s classifier loses about **80% of its profit** (38.6% → 7.3% gross) when the entry moves from the pre-release price to one minute after the release; its Sharpe falls from about 2.1 to 0.9. The first minute is where the market prices the news, and it cannot be traded. Jack’s independent rebuild of Aaryen’s model found the same (about 96% of profit per trade lost).
- **Costs explain the second largest.** The high-turnover DTW strategies (300+ trades a year) are positive only at ¼ tick per side; at 2 ticks + \$4.50 they lose about 20%. The surprise strategy trades a third as often with more edge per trade, so it survives realistic costs.
- **At equal costs and an executable entry, the macro-surprise book has the highest total return in the team at every cost level, and the highest Sharpe at gross and ¼-tick costs.** At our 2-tick cost, Aaryen’s executable 30-minute classifier has a similar or slightly higher Sharpe (monthly 0.84 vs 0.72 for ES + NQ), but with far fewer trades (55 vs 195 a year) and a much smaller total return (5.6% vs 25%). The two are built on related inputs (both use the macro surprise), which supports the conclusion that the surprise is the source of the executable edge.

7. Discussion

7.1 Why does a drift exist in equity futures, and not in bonds?

Figure 2 and Sections 6.2–6.3 show the same pattern: a large jump at the release, then a drift in the same direction for about 30 minutes in ES and NQ, and almost none in ZN. Several explanations are consistent with the evidence:

- **Interpretation takes time for equities.** A CPI print maps into bond prices almost mechanically (expected policy rate → yield). For equities, investors must also judge the effect on growth, margins and risk premia, and large asset managers rebalance over minutes rather than milliseconds. The fact that the drift is stronger in **NQ**, whose value depends more on long-dated cash flows, fits this channel.
- **Liquidity after the release.** Order books are thin in the seconds after a release and refill over the following minutes. Part of the drift may be price discovery as liquidity returns.
- **The bond market is the first mover.** Treasury futures are where the rate information is priced directly and by the fastest participants. By the time a strategy can enter ZN, the move is complete.
- **Costs relative to moves.** Even with some drift, ZN’s cost (3 bps per round trip) is large relative to its post-release moves (mean |30-min drift| 10.5 bps for CPI). ES and NQ have larger moves and, in NQ’s case, a cost below 1 bps.

7.2 The edge is regime-dependent

Around 75–80% of the CPI strategy’s profits came from 2022, when inflation was the market’s dominant concern and every CPI print moved the expected Fed path. The multi-release book spreads its risk across ~10 releases and depends less on one regime (Sharpe ex-2022: 0.28 in ES, 0.44 in NQ), but it is also strongest in 2022–2026. The practical reading is that **macro-surprise trading pays when macro data drive policy expectations**: a live implementation should scale up when inflation or labour data are the market’s main theme and down when they are not.

7.3 The pre-event state: size, not direction

The state results are a clear example of the difference between **in-sample explanation** and **out-of-sample trading**:

- In sample (Notebook 03), volatility clearly changes both the size of the move and the CPI surprise beta in ES and NQ.
- Out of sample (Notebook 04), no state model beats surprise only, because (i) the effect is mostly on the untradeable first-minute jump, (ii) it became detectable in real time only in 2022, and (iii) the “best” state changed from year to year.

The state is still valuable for **risk management**: in high-volatility states the 30-minute move is about 20 bps larger, so position sizes and stop-losses should be scaled by volatility even though the direction comes from the surprise.

7.4 DTW: no directional information at an executable entry

DTW was tested more thoroughly here than in any other part of the team’s work: three markets, five windows, four combinations, with the peer’s exact specification and a validation on synthetic data. The rank IC is 0.01–0.04 everywhere, and adding DTW to the surprise lowers the Sharpe in every equity-market combination. This agrees with Jack Duncan’s own conclusion that DTW direction is “not established”, and extends it: DTW does not work as a **confirmation** of the surprise either. The DTW path shape is informative about **volatility** (dispersion), not about **direction**.

7.5 The trade-count question

The supervisor’s question, “can the strategy trade as often as the peer’s ~185 times a year?”, has a precise answer:

Route to more trades	Trades / yr	Sharpe	Verdict
CPI only (ES)	9–12	0.76–1.03	Strong but rare
~10 releases, one window (ES)	98	0.65	Best single-market book
~10 releases, one window (NQ)	97	0.71	Best single-market book
~10 releases, ES + NQ	195	0.72	Reaches the peer’s trade count with a higher Sharpe
Five windows per release (ES)	314–490	–0.75 to –1.23	Adds only losing trades
Adding ZN	277	–0.75	Adds only losing trades

The right way to trade more is **more markets with a real drift (ES, NQ), not more windows per release or markets without a drift**.

7.6 Execution: the decisive assumption

Two execution assumptions drive most of the differences between the team’s reported numbers:

1. **Entry time.** Entering at the pre-release price captures the jump, which no real strategy can. It roughly doubles a strategy's Sharpe (Aaryen: 2.1 $\rightarrow$ 0.9).
2. **Costs.** $\frac{1}{4}$ tick per side (the peer assumption) is plausible for patient limit orders in normal markets, but not for market orders in the first minute after a release, when spreads widen. At 2 ticks + \$4.50, strategies with 300+ trades a year lose money; the surprise book, with more edge per trade, survives.

A paper-trading test with real fills in the first minute after a release is the natural next step before any live use.

7.7 Limitations

- **Sample size.** About 120 releases per family (86 CPI releases in the evaluation window); CPI-only results rest on 50–90 trades.
- **One regime dominates.** The 2022 inflation shock drives much of the CPI result.
- **Surprise definition.** For near-zero surprises the sign depends on which CPI line is used (Section 6.5.4). A fixed definition, or skipping $|z| < 0.5$, should be tested and labelled as a post-hoc refinement.
- **Impact screen.** It admits families with no real surprise (FOMC decision, interest on reserve balances); excluding them is a simple improvement.
- **Consensus data.** Bloomberg survey medians can be revised or stale for thin lines; the $\pm 5\sigma$ cap and the 24-release minimum limit the damage.
- **Costs and fills.** Costs are modelled, not observed. First-minute execution may be worse than 2 ticks.
- **Team replication.** Jack's strategy is rebuilt from its specification with our data and family grouping; the rebuild (310 trades a year, Sharpe 0.16) does not reproduce his slide (185, 0.61), so the Notebook 08 comparison of his strategy is approximate.
- **NQ and ZN data.** Jack's cleaned files include forward-filled minutes; 9 NQ and 13 ZN releases lack price data around the release.

8. Conclusion and recommendations

8.1 Answers to the research questions

RQ1. Which macro releases move ES, NQ and ZN, and by how much per unit of surprise?

Short answer: CPI, in all three markets. Other releases move markets, but their surprises explain little of the direction.

Evidence. - A one-standard-deviation hotter headline CPI print lowers **ES by about 21 bps, NQ by about 28 bps and ZN by about 10 bps** within five minutes ($t \approx -2.7$ to -2.8 in every market). The surprise explains **27–30%** of the five-minute reaction in all three (Section 6.2, Figures 3–6). - In ES the CPI coefficient is stable at -18 to -24 bps from 1 to 60 minutes and significant at 1% at every horizon. - The **Employment Report** moves ES almost as much as CPI (mean |30-min reaction| 30 bps against 38 bps) but its surprise explains only 3–7% of the direction. Payrolls, the unemployment rate and wages often point in different directions, and the market weighs stronger growth against higher rates. - **PCE** moves markets less (15 bps in ES), because most of its information is already known from CPI and PPI earlier in the month.

Why it matters. CPI is the release with both a large move and a predictable direction. It is the natural core of any macro-surprise strategy, and the rest of the study is built around it.

RQ2. How much of the reaction happens in the first minute, and how much remains as a tradeable drift?

Short answer: most of it happens in the first minute. In ES and NQ a drift of roughly a quarter of the CPI reaction remains for about 30 minutes; in ZN almost nothing remains.

Evidence. - CPI reaction per 1-SD surprise: about **35 bps in ES and 37 bps in NQ at five minutes**, against a 30-minute drift after the first minute of about **9 bps (ES) and 8.5 bps (NQ)**. In ZN the five-minute reaction is 14 bps but the drift is only **0.8 bps** (Section 6.3.1). - Figure 2 shows the same thing visually: a jump at the release, then a slower continuation in ES and NQ. - The five-window ladder (Section 6.6.3) confirms the timing: the surprise's rank IC is 0.14 (ES) and 0.09 (NQ) in the first 30 minutes and about zero in every later window.

Why it matters. A strategy that reacts one minute after the release can only trade the drift. In ES and NQ the drift is several times larger than the cost (1.7 and 0.7 bps); in ZN it is smaller than the cost (3 bps). This single comparison explains most of the trading results.

RQ3. Does the market's state before the release change the reaction to the same surprise?

Short answer: yes for the **size** of the move in equity markets, and in sample also for the CPI response; but the effect is mostly on the untradeable jump and mostly after 2021.

Evidence. - **Size:** 198 of 360 magnitude tests survive FDR correction in ES and 154 in NQ. In high-volatility states the absolute 30-minute move is about 18–20 bps larger ($t \approx 4$ –5). None survive in ZN (Section 6.3.5). - **Direction (surprise beta):** the CPI beta rises from about 11–18 bps in calm markets to 50–70 bps in volatile markets, in both ES and NQ (Figures 9–10). The interaction is $t = 3.8$ in ES and $t = 3.0$ in NQ, but it survives FDR only in ES, is strongest on the first-minute reaction (Figure 12), and is significant only after 2021. - **ZN:** the only robust state effect is negative and small: after a volatile day the post-release drift follows the surprise less ($t = -3.2$, -4.6 bps across the full range). It is the only state effect that is significant in both subperiods.

Why it matters. The state is a good predictor of **risk** (how much the market will move) but a weak and unstable predictor of **return** (which way it will drift).

RQ4. Can a trader profit in real time, after costs, with only past information? Does the state help?

Short answer: yes in ES and NQ, no in ZN; and no, the state does not help out of sample.

Evidence. - **CPI, 2019–2026, net of 2 ticks + \$4.50:** Sharpe **1.03 (naive) and 0.99 (surprise only) in ES, 1.00 and 0.79 in NQ**. The ES surprise-only Sharpe has a 90% bootstrap interval of 0.25–1.72; it survives 8 ticks of cost (0.62) and a 5-minute entry delay (0.55–0.71) (Section 6.4, Figures 14–22). - **ZN:** the naive rule earns +2.7 bps per trade before costs (gross Sharpe 0.53) but -0.3 bps after the 3 bps cost; the fitted models trade only 1–5 times in 7.5 years. - **State:** no state model beats surprise only in any market (ES: -0.21 to $+0.08$ Sharpe; NQ: -0.05 to $+0.03$; none significant). The state effect became detectable in real time only in 2022 (Figure 17), and the adaptive model chose a different state in different years. - **Regime:** 74–79% of CPI profits come from 2022; excluding 2022 the Sharpe is 0.3–0.7.

Why it matters. The economic direction of the surprise is enough; estimating a more complex model adds noise. In NQ the naive rule is even significantly better than the fitted model, because with sub-1-bp costs it pays to trade every CPI release.

RQ5. How many trades a year can a surprise strategy make across many releases, and does the edge survive?

Short answer: about 100 trades a year per market, with a positive and significant edge in ES and NQ.

Evidence. - The multi-release naive book (~ 10 screened families) makes **98 trades a year in ES (Sharpe 0.65) and 97 in NQ (Sharpe 0.71)**, with random-sign p-values below 0.001 and 0.005 (Section 6.5). - It depends less on 2022 than CPI alone (Sharpe ex-2022: 0.28 in ES, 0.44 in NQ). - Its weakness is cost: in ES its Sharpe falls to 0.20 at 4 ticks, while NQ keeps 0.60. - More sophisticated sizing does not help: per-family surprise models (0.11–0.37) and volatility sizing (0.37–0.56) are below the naive book. - A second window after each release loses money in every market (ES -0.79 , NQ -0.23 , ZN -3.08).

Why it matters. The trade count can be raised tenfold without losing the edge, but only by adding **releases** (and markets), not by trading longer after each release.

RQ6. Does the DTW pattern add information? Can more post-release windows raise the trade count profitably?

Short answer: no, and no.

Evidence. - DTW's rank IC with the 30-minute return is **0.01 (ES), 0.03 (NQ) and 0.04 (ZN)**, with hit rates of 47–51% (Section 6.6.1). - Adding DTW lowers the multi-release Sharpe in ES (0.65 → 0.31–0.41) and NQ (0.71 → 0.62–0.67). The CPI-only results favour different combinations in ES and NQ and none is significant (Figure 27). - The five-window ladder raises the trade count to 270–490 a year, above the peer's 185, but lowers the Sharpe in every market (ES –0.75 to –1.23; NQ +0.27 at best; ZN –3.9 to –7.7) (Figures 28–29).

Why it matters. The shape of the price path carries information about volatility, not direction, and the surprise's information is used up within 30 minutes. Neither route adds profitable trades.

RQ7. Does combining markets help, and how do the team's workstreams compare at equal costs?

Short answer: ES + NQ combine well; ZN does not. At equal costs and executable entries, the surprise book has the highest total return in the team.

Evidence. - **ES + NQ:** 195 trades a year, Sharpe **0.72**, total +25.2%, drawdown –7.4%; at ¼ tick per side, Sharpe 1.01 and total +35.8% (Section 6.7). - **Adding ZN:** the three-market book loses (Sharpe –0.75), because ZN has no edge and receives the largest risk weight. - **Team comparison:** at ¼ tick per side, the ES surprise book has a daily Sharpe of 1.12 (total 21.1%), Aaryen's executable classifier 0.88 (6.9%), Hoshea's DTW factor 0.25 (10.3%), and the rebuild of Jack's ladder 0.07 (2.4%). At 2 ticks + \$4.50 the two DTW strategies lose about 20%. Aaryen's reported Sharpe of about 2.1 falls to about 0.9 once the entry is moved one minute after the release (Section 6.8).

Why it matters. The two decisive assumptions in the team's work are **entry time** and **cost**. Once they are equal, the macro surprise is the most robust source of return.

8.2 Final model comparison

Model family	Best result (net, 2019–2026)	Verdict
Surprise only	CPI: Sharpe ~1.0 (ES); ~10 releases: 0.65 (ES), 0.71 (NQ); ES + NQ: 0.72 at 195 trades a year	Recommended
Surprise + State	CPI: 0.77 (ES), 0.75 (NQ)	Worse out of sample; use the state for risk sizing only
Surprise + Pattern (DTW)	~10 releases: 0.31–0.41 (ES), 0.62–0.67 (NQ)	Worse; DTW has no directional information
Ladder (5 windows)	–0.75 to –1.23 (ES)	Adds only losing trades
Any ZN strategy	–0.3 to –2.4	Not tradeable at realistic costs

8.3 Recommendations

The recommendations are ordered from the core trading decision to implementation and governance.

1. Trade the macro surprise only in equity-index futures (ES and NQ). - The drift after the first minute is several times the cost in ES and NQ and smaller than the cost in ZN. - NQ should be treated as an equal partner to ES, not an add-on: its multi-release book is the strongest single result of the study (Sharpe 0.71, 0.44 excluding 2022, 0.60 at 4 ticks).

2. Use one simple, pre-specified trade rule. - Entry at the close of the first post-release minute; exit 30 minutes after the release. - Direction = sign of the family surprise x , with line signs re-estimated once a year from the historical jump. - No second window, no DTW filter, no state-dependent direction. Every one of these was tested and lowered the Sharpe.

3. Build a two-leg book. - **CPI core leg:** the CPI surprise model or signal $\times$ volatility sizing (Sharpe 0.74–0.77 in ES and NQ, 8–16 bps per trade, robust to $4\times$ costs). This leg carries most of the edge per trade. - **Multi-release leg:** the naive sign on the ~ 10 families admitted each year by the impact screen, in both ES and NQ (~ 195 trades a year combined). This leg carries diversification and reduces dependence on 2022. - Allocate risk between the legs so that neither exceeds about two-thirds of the book's expected volatility.

4. Use the pre-event state for risk, not direction. - Scale the position inversely to the expected size of the move: in high rv_{20d} states the 30-minute move is about 20 bps larger, so a volatility-targeted position (for example, contracts $\propto 1 / \text{recent 30-minute move size}$) keeps the risk per trade constant. - Set stop-losses and daily loss limits in units of expected move, not fixed points.

5. Clean the release universe. - Exclude families with no genuine surprise from the impact screen (FOMC rate decision, interest on reserve balances): they rank high because they move markets, but they have almost nothing to trade. - Fix one CPI surprise definition in advance (for example headline CPI MoM, or the average of headline and core MoM) and do not trade CPI when $|z| < 0.5$, where the sign depends on the definition.

6. Scale exposure with the macro regime. - The strategy earned most when macro data drove policy expectations (2022–2023). - A simple, pre-specified regime indicator, for example the size of recent CPI reactions or the correlation between CPI surprises and 2-year yield changes, can scale the book between a base and a maximum allocation.

7. Protect the execution assumption. - The ES multi-release book loses most of its edge above 4 ticks of round-trip slippage. Use limit or pegged orders with a short time-out rather than market orders at the release, and do not trade if the spread one minute after the release is wider than a set threshold. - Paper-trade for at least three to six months and compare real fills with the modelled 2 ticks before allocating capital.

8. Monitor the edge continuously. - Track a rolling 12-month rank IC of the surprise with the 30-minute drift, the rolling Sharpe, and the realised cost per trade. - Pre-specify a stop rule: for example, reduce the allocation by half if the rolling IC is below zero for 12 months, and stop if the rolling Sharpe is below -0.5 over 24 months.

9. For the team. - Report every strategy with an **executable entry** (after the release) and at **two cost levels** ($\frac{1}{4}$ tick per side and 2 ticks + \$4.50) on the same 2019–2026 window. Notebook 08 shows that these two choices change Sharpe ratios by more than any model choice.

8.4 Future work

1. A pre-registered, fixed CPI surprise definition - *Motivation:* the CPI naive book's Sharpe changes from 1.03 to 0.51 depending on which CPI lines define the surprise (Section 6.5.4). - *Method:* choose one definition (headline MoM, core MoM, or their average) and a minimum $|z|$ threshold before looking at 2019–2026 results; evaluate it on a fresh hold-out period (2026 onwards). - *Expected outcome:* a more stable CPI leg, and a clean measure of how much of the Notebook 04 result depended on near-zero surprises.

2. A cleaner impact screen - *Motivation:* the screen admits families with no real surprise (FOMC, interest on reserves) and its rankings beyond CPI are unstable across markets. - *Method:* require a minimum share of releases with a consensus and a minimum historical surprise-jump correlation before a family is admitted; compare the resulting books with the current ones walk-forward. - *Expected outcome:* fewer wasted trades and a higher edge per trade in the multi-release leg.

3. Realistic execution with order-book data - *Motivation:* costs are the most important untested assumption, especially for the ES multi-release book. - *Method:* use Level-2 (market-by-price) data around releases to measure spreads, depth and slippage in the first minutes; simulate limit-order

entries with fill probabilities; test entries at 10, 20, 30 and 60 seconds after the release. - *Expected outcome*: a measured cost per trade by release and market, and possibly an earlier entry that captures more of the drift.

4. Paper trading with a live release feed - *Motivation*: the backtest assumes the actual value and consensus are known instantly and correctly. - *Method*: connect a machine-readable release feed (for example Bloomberg ECO or a low-latency news feed), run the strategy in simulation for several months, and log decision latency, data errors and fills. - *Expected outcome*: a realistic estimate of implementation shortfall and operational risk before any capital is allocated.

5. More equity-index markets - *Motivation*: the edge exists in both equity indices tested; adding markets is the only route to more trades that did not lower the Sharpe. - *Method*: apply the same pipeline to Russell 2000 (RTY), Dow (YM), Euro Stoxx 50 and Nikkei futures around US releases, with market-specific costs and walk-forward signs. - *Expected outcome*: a broader book with more trades a year, and a test of whether the drift is a US-market or a global effect.

6. Rates markets closer to policy expectations - *Motivation*: ZN prices the surprise in the first minute; shorter-maturity contracts that respond directly to policy expectations may behave differently. - *Method*: repeat Notebooks 02b-05b on 2-year Treasury futures (ZT), SOFR futures and the Fed-funds curve, measuring jump and drift separately. - *Expected outcome*: either a rates market with a tradeable drift, or confirmation that rates markets are efficient within the first minute.

7. Volatility and risk forecasting - *Motivation*: the pre-event state and the DTW dispersion predict the **size** of post-release moves well, even though neither predicts direction. - *Method*: build a model of the expected absolute 30-minute move from the state variables and DTW dispersion; use it for position sizing, stop-loss levels, and possibly options strategies around releases. - *Expected outcome*: a lower drawdown for the same expected return, and a potential second strategy that trades volatility rather than direction.

8. Richer surprise information - *Motivation*: the headline surprise ignores revisions, the distribution of forecasts, and the text of releases. - *Method*: add revisions to previous months, the dispersion of economists' forecasts (a surprise is more informative when forecasters agree), and the surprise relative to "whisper" numbers or market-implied expectations (for example CPI swaps). - *Expected outcome*: a sharper signal for the releases where the headline alone is ambiguous, especially the Employment Report.

9. Regime-aware allocation - *Motivation*: 75-80% of CPI profits came from 2022, when inflation dominated policy expectations. - *Method*: test pre-specified regime indicators (recent size of CPI reactions, the correlation of surprises with 2-year yields, the level of inflation relative to target) as scaling factors, strictly walk-forward. - *Expected outcome*: higher risk when the macro channel is active and lower risk when it is not, with a smaller drawdown in quiet regimes.

10. A longer and independent sample - *Motivation*: the evaluation window (2019-2026) contains one dominant inflation regime and about 86 CPI releases. - *Method*: extend the event study back to 2000 with daily or tick data where available, and keep 2026 onwards as an untouched out-of-sample period. - *Expected outcome*: a more reliable estimate of the long-run edge and its variation across monetary-policy regimes.

Overall: across three model families (macro surprise, pre-event state, DTW patterns) and three markets (ES, NQ, ZN), only the **macro surprise in the first 30 minutes after a release, in equity-index futures**, gives a robust, tradeable directional edge after realistic costs. Combining ES and NQ gives about 195 trades a year at a net Sharpe of about 0.72, with an economically explained signal.

9. Reproducibility

9.1 Data placement (not in the repository)

File	Location
ES_1m_2010_2026_full.parquet (Databento, with instrument_id)	Reza/data/raw/
Bloomberg Economic Releases.xlsx	Reza/data/raw/
futures_1min_clean_v3_NQ.parquet,	the repository's data/processed/
futures_1min_clean_v3_ZN.parquet (Jack Duncan)	
Hoshea's dtw_simple_generalized_best_factor.parquet,	Reza/data/team_inputs/
Aaryen's	
aaryen_backtest_predictions_{10,30,60}min.parquet	
(Notebook 08 only)	

9.2 Run order

01 → 02 → 02b

03 → 04 → 05 → 06 (ES)

03b → 04b → 05b → 06b for NQ, then for ZN

07 → 08

scripts/make_readme_figures.py (Figures 1 and 2 of this document)

Each notebook reads the previous one's outputs from data/processed/ and results/ and checks that it reproduces them (for example, Notebook 05 reproduces Notebook 04's CPI trade returns to within 10^{-14} bps). The b-notebooks save everything as nb0Xb_{MARKET}_*, so ES outputs are never overwritten.

9.3 Environment

Python 3.10+, numpy, pandas, matplotlib, pyarrow (parquet), openpyxl (Bloomberg workbook). No statsmodels or DTW library is required; HC3 OLS, the statistical tests and DTW are implemented in src/. Notebook 06's DTW runs in a few seconds per market; the five-window ladder in about 20 seconds.

9.4 Repository layout

```

Reza/
├── README.md                this document
├── notebooks/
│   ├── 01_macro_surprise_cleaning.ipynb
│   ├── 02_event_response_analysis.ipynb
│   ├── 02b_event_response_NQ_ZN.ipynb
│   ├── 03_state_conditioning.ipynb          03b_state_conditioning_{NQ,ZN}.ipynb
│   ├── 04_walk_forward_strategy.ipynb       04b_walk_forward_strategy_{NQ,ZN}.ipynb
│   ├── 05_scaling_surprise_strategy.ipynb   05b_scaling_surprise_strategy_{NQ,ZN}.ipynb
│   ├── 06_surprise_plus_dtw.ipynb           06b_surprise_plus_dtw_{NQ,ZN}.ipynb
│   ├── 07_multi_market_surprise_dtw.ipynb
│   └── 08_team_comparison_equal_costs.ipynb
├── src/                          state_conditioning, walk_forward, surprise_universe,
│                                   dtw_signal, multi_asset, team_replication
├── scripts/make_readme_figures.py
├── results/                      CSV outputs, prefixed by notebook (nb03_, nb04b_NQ_, ...)
├── figures/                      PNG outputs, same prefixes
└── data/                        raw/ and processed/ (gitignored)

```

Every notebook ends with a **Purpose, Research Questions, Answers and Conclusion** section that documents its results in detail.

10. Appendix

10.1 Key result files

File	Content
results/final_macro_event_comparison.csv	Notebook 02 summary: impact and directional rank of CPI, NFP, PCE
results/nb02b_summary.csv	Reaction and drift betas for ES, NQ, ZN
results/nb03_interaction_results.csv, nb03b_{NQ,ZN}_...	All 360 state interaction tests per market
results/nb03_state_selection.csv	The four-criterion selection
results/nb04_final_comparison.csv, nb04b_{NQ,ZN}_...	CPI walk-forward strategies
results/nb04_trade_log.csv	Every walk-forward trade
results/nb05_final_comparison.csv, nb05b_{NQ,ZN}_...	Multi-release books
results/nb06_final_three_model_comparison.csv	Surprise vs state vs pattern
results/nb06_ladder_books.csv	Five-window ladder
results/nb07_final_comparison.csv, nb07_portfolio.csv	Multi-market portfolio
results/nb08_team_comparison_equal_costs.csv	All team workstreams, four cost scenarios
results/nb08_slide_format_by_person.csv	Slide-format tables per person

10.2 More figures

Every notebook saves its figures in figures/. The NQ and ZN versions of every ES figure carry the nb0Xb_NQ_ and nb0Xb_ZN_ prefixes, for example:

ES	NQ	ZN
nb03_interaction_tstats_CPI.png	nb03b_NQ_interaction_tstats_CPI.png	nb03b_ZN_interaction_tstats_CPI.png
nb04_latency_holding_grid.png	nb04b_NQ_latency_holding_grid.png	nb04b_ZN_latency_holding_grid.png
nb05_family_contribution.png	nb05b_NQ_family_contribution.png	nb05b_ZN_family_contribution.png
nb06_ladder_sharpe_by_window.png	nb06b_NQ_ladder_sharpe_by_window.png	nb06b_ZN_ladder_sharpe_by_window.png

10.3 Glossary

Term	Meaning
Surprise	Actual minus Bloomberg consensus, standardised by the standard deviation of previous surprises
Jump	Return from the last pre-release price to the close of the release minute; not tradeable
Drift	Return from the close of the release minute ($\approx T+1$) to $T+30$; what the strategies trade
Naive sign	Trade the sign of the surprise, one contract, no estimation
Walk-forward	Every parameter estimated only from releases before the one being traded
Rank IC	Spearman correlation between a signal and the subsequent return
Random-sign test	Share of 1,000 books with random directions (same trades and costs) that match the strategy's Sharpe
BH-FDR	Benjamini-Hochberg control of the false discovery rate across many tests
DTW	Dynamic Time Warping; a distance between two paths that allows local stretching in time
Ladder	Five consecutive 30-minute trades after each release ($T+1$, $T+31$, $T+61$, $T+91$, $T+121$)
Tick	Minimum price step: 0.25 points (ES, NQ), 1/64 point (ZN)

10.4 Credits

- **Jack Duncan:** the release-family rule, impact screen, volatility sizing, DTW specification and ladder, contract specifications, and the cleaned ES/NQ/ZN 1-minute data.
- **Hoshea Zeng:** the generalised DTW “event game” factor.
- **Aaryen Mehta:** the $\pm 5\sigma$ winsorisation and the surprise + DTW + momentum classifier.

- **Reza Zamani:** causal surprises, event studies, state conditioning, the walk-forward engine, the multi-release and multi-market books, the surprise + DTW tests, and the team comparison at equal costs.

11. Acknowledgements

This project was carried out as part of the UC Berkeley Haas Master of Financial Engineering (MFE) industry project program.

- **Benjamin Steel (BlackRock)** proposed the research question and supervised the project, including the suggestion to extend the study from ES to NQ and ZN and to compare strategies by trade frequency.
- **Ian Kaufman and Chris Pohalski (UC Berkeley Haas, MFE program)** coordinated the partnership between the team and BlackRock and organized and attended the project meetings.

We also thank our teammates Jack Duncan, Hoshea Zeng and Aaryen Mehta, whose work on DTW strategies, release screening and surprise classification is credited throughout this document.

Disclaimer. This is an industry project carried out by MFE students. The views, methods and results are the authors' own and do not represent the views of BlackRock or UC Berkeley. Nothing in this document is investment advice. Market data used in the study are proprietary and are not included in the repository.